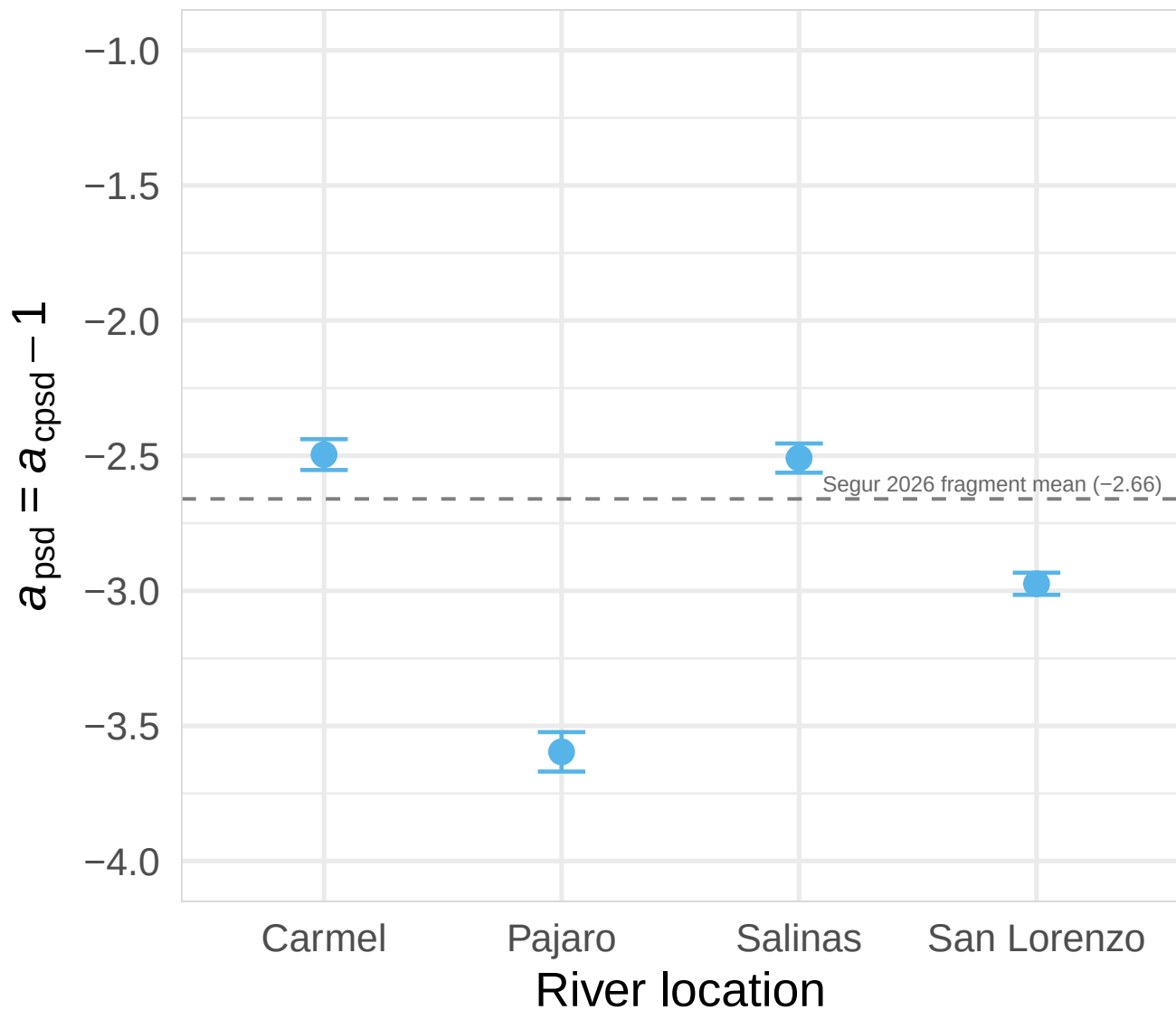


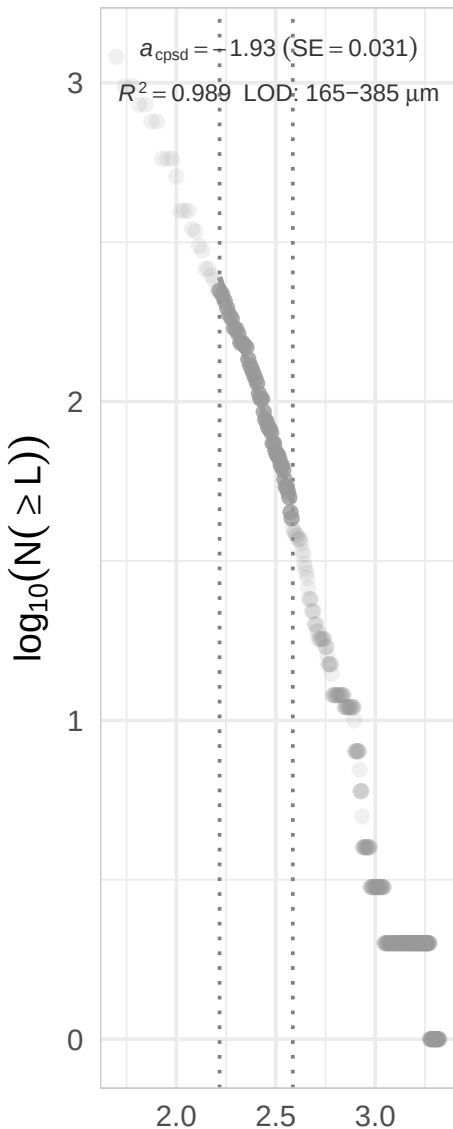
# Power-law slope by river location

Error bars =  $\pm 1$  SE | Dashed = Segur 2026 reference

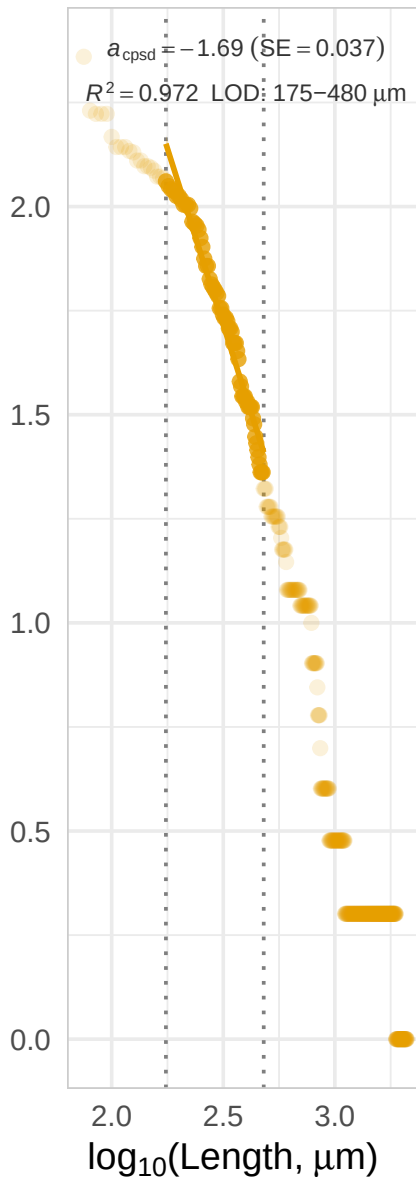


# C-PSD length fits by shape

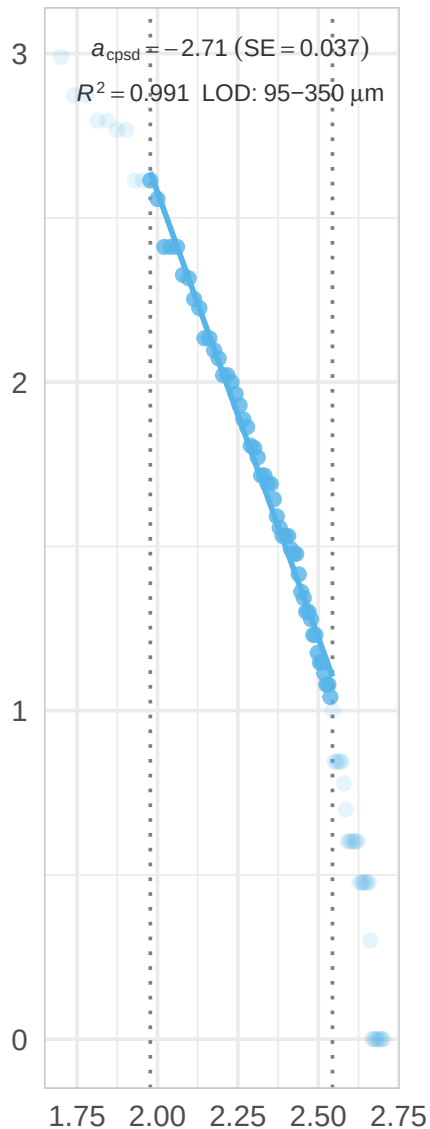
**all**



**fiber**

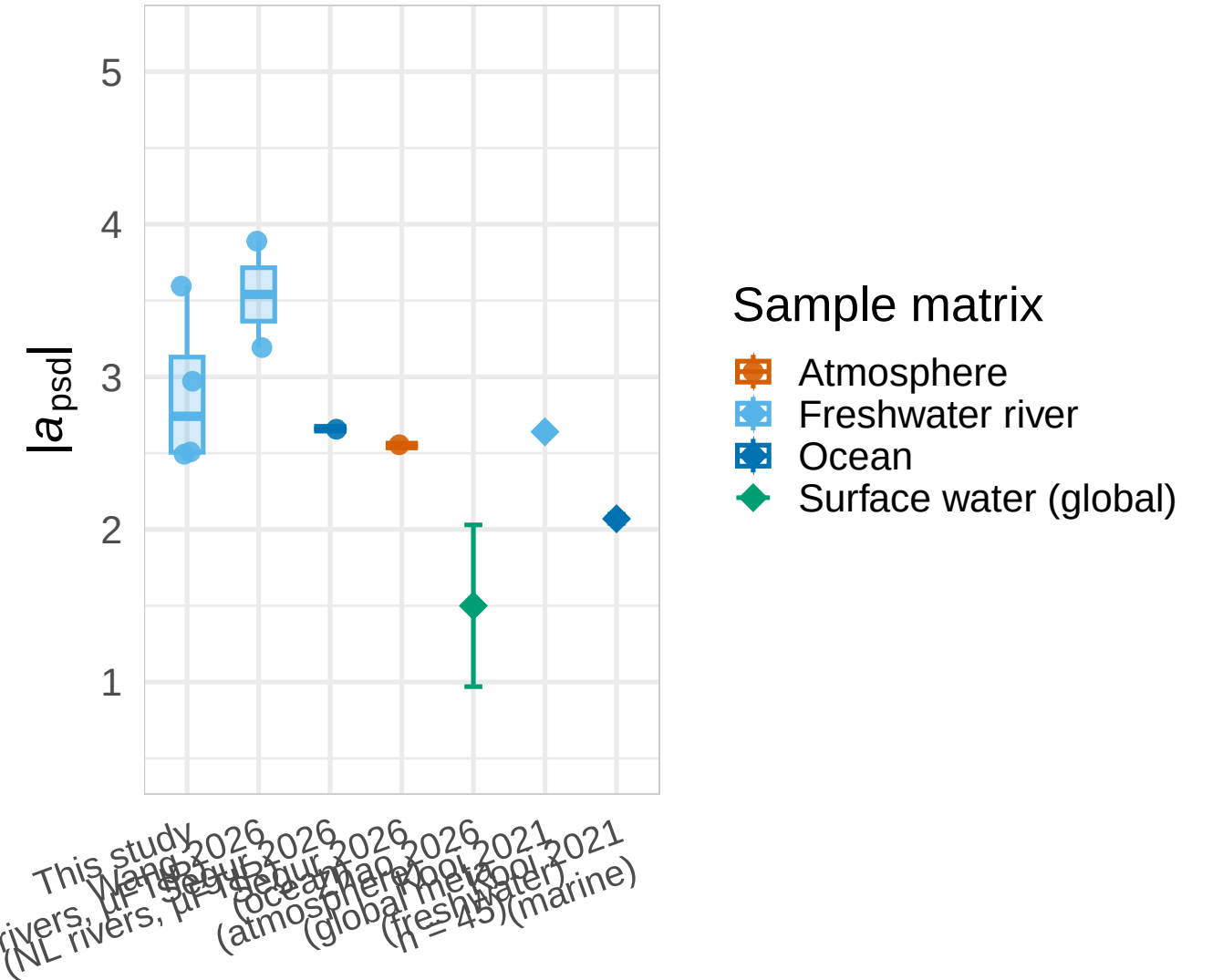


**fragment**



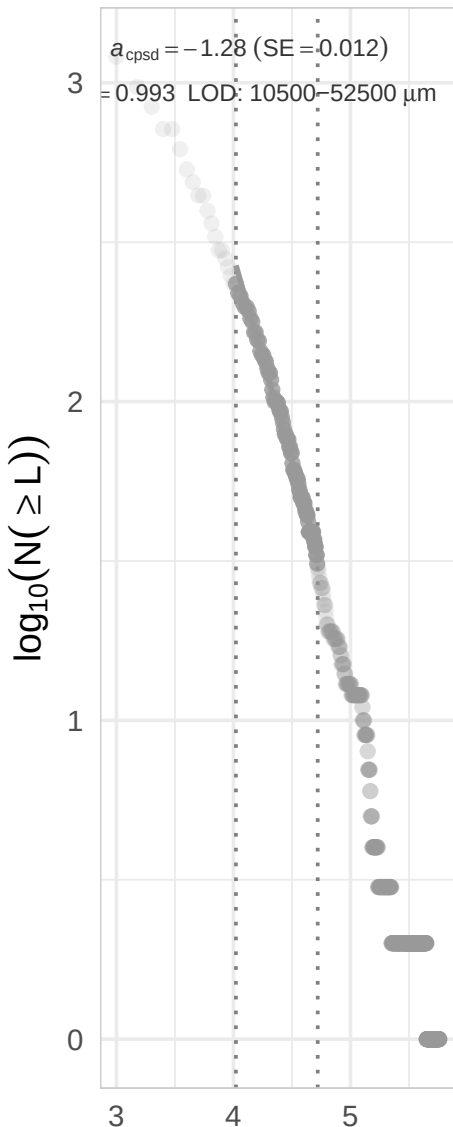
# Power-law size distribution slope: this study

Boxes = IQR (per-site values). Diamond = mean; circles = outliers

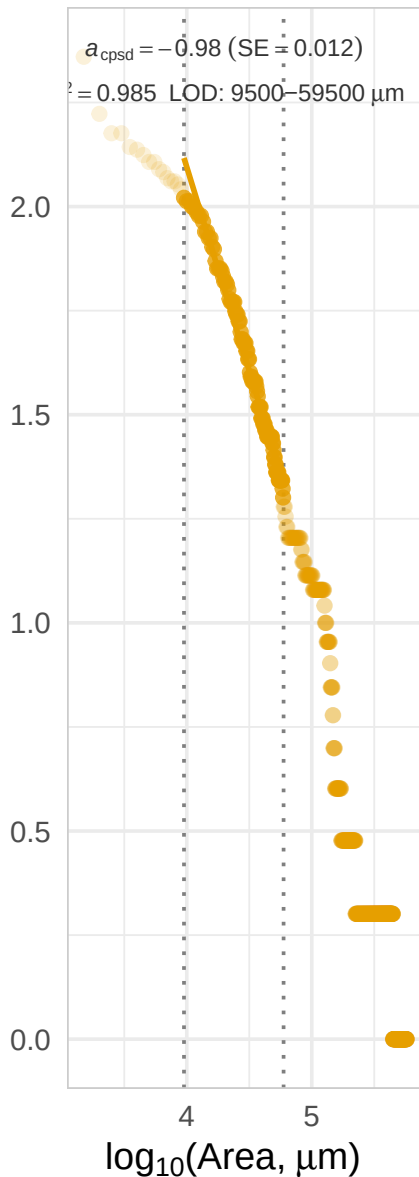


# C-PSD fits by shape

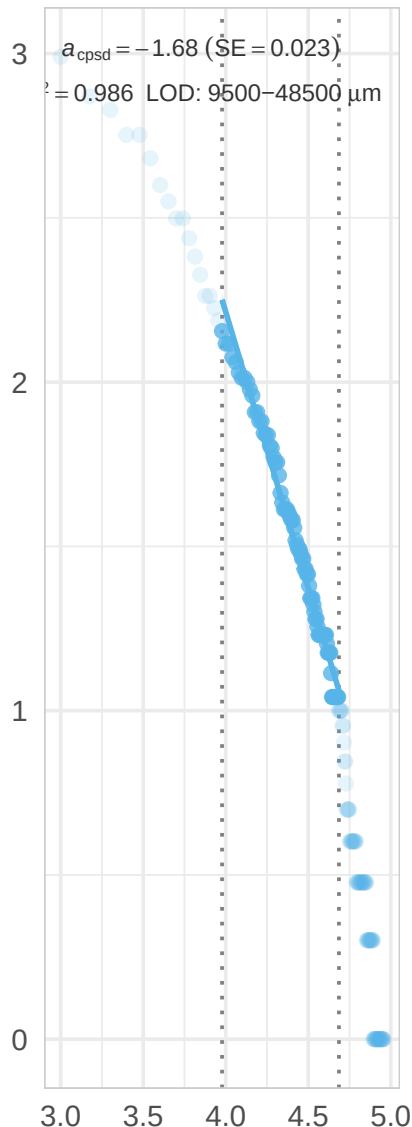
**all**



**fiber**



**fragment**

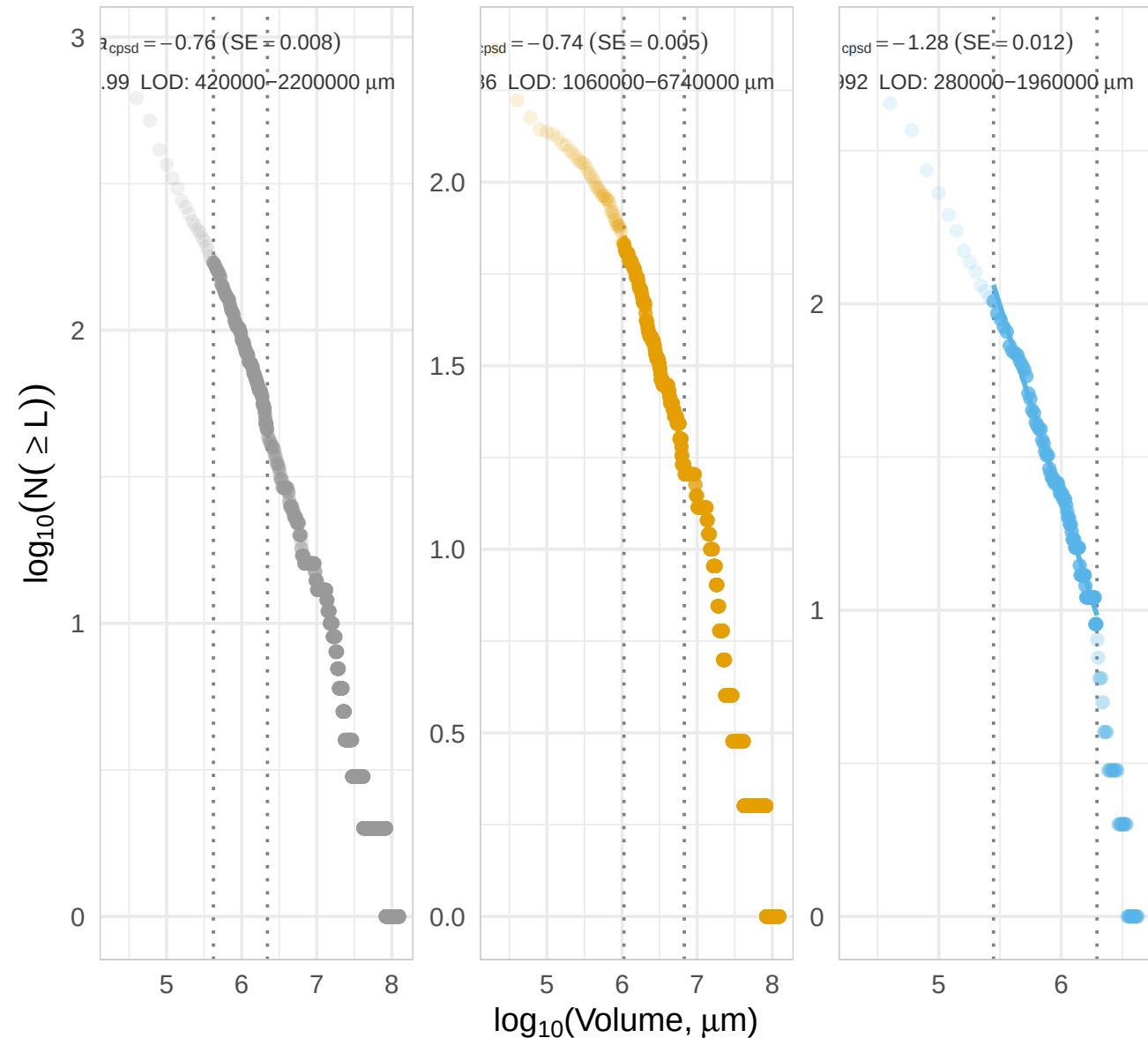


# C-PSD fits by shape

**all**

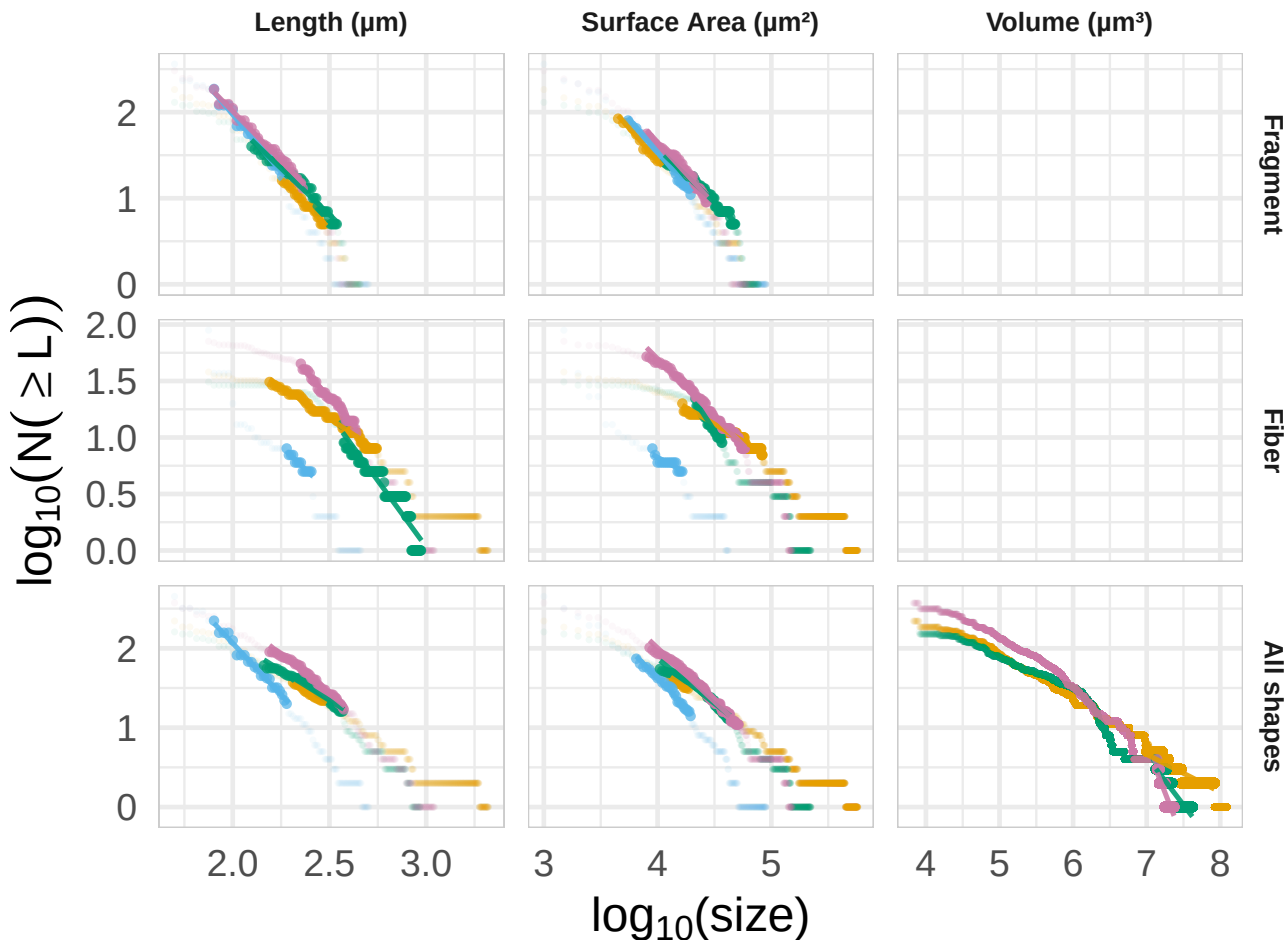
**fiber**

**fragment**



# C-PSD power-law fits by river, shape, and size

Solid lines + ribbon = fit  $\pm$  95% CI within Segur LOD window

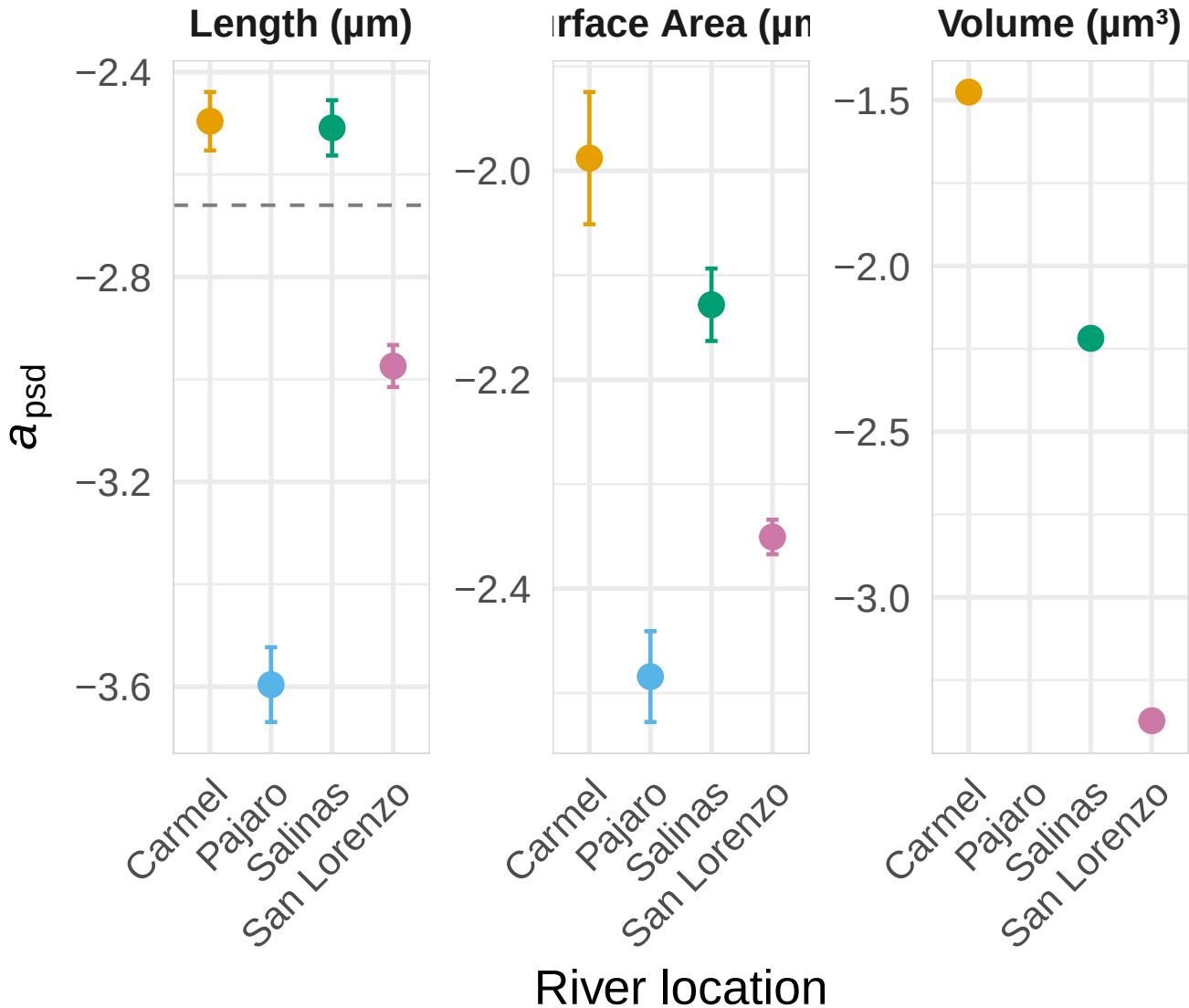


River location — Carmel — Pajaro — Salinas — San Lorenzo

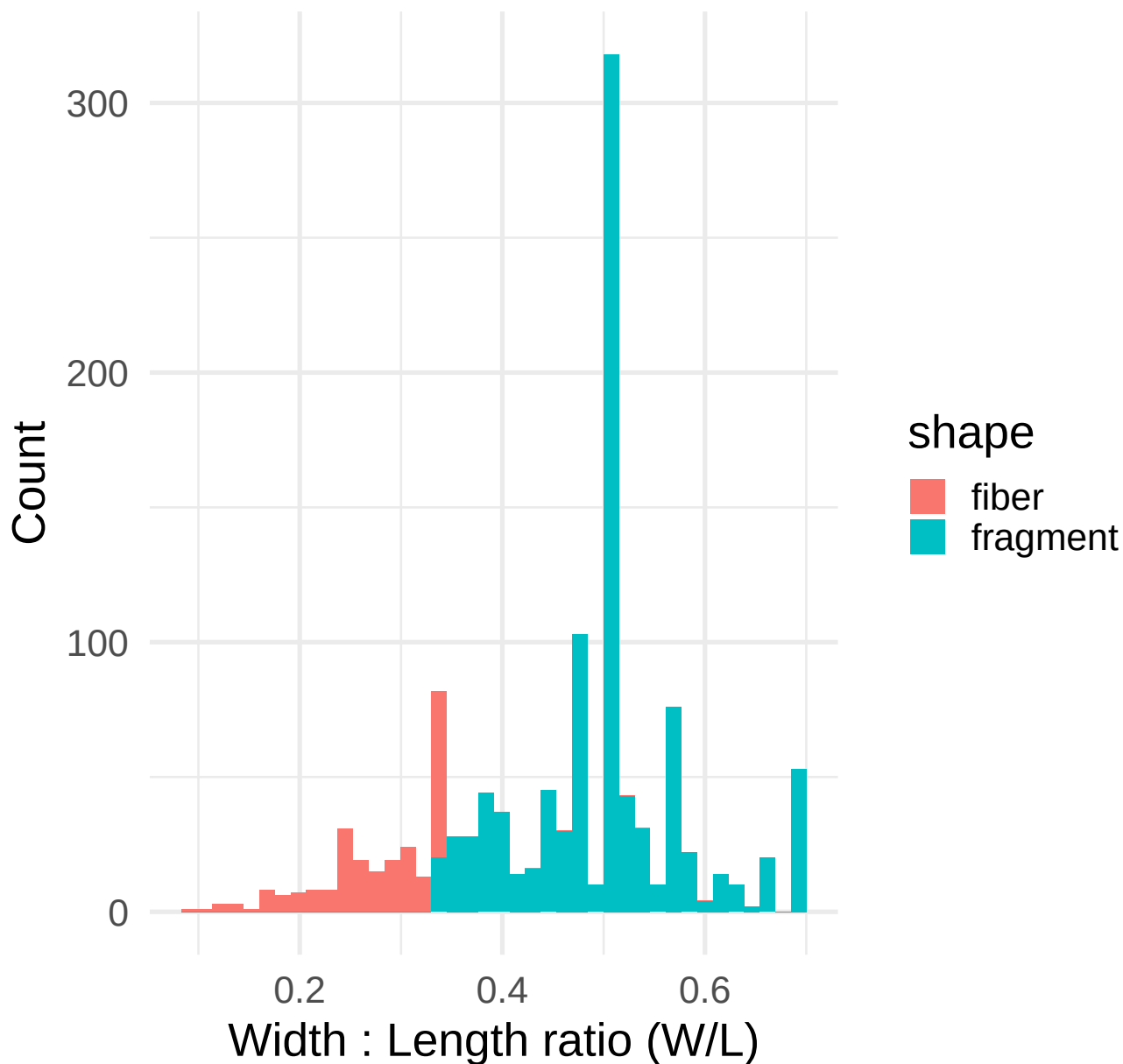
Rows: shape. Columns: size metric. Fits per river via Segur (2026) LOD algorithm.

# BN-PSD slope by river location and size

Error bars =  $\pm 1$  SE | Dashed = Segur 2026 frag



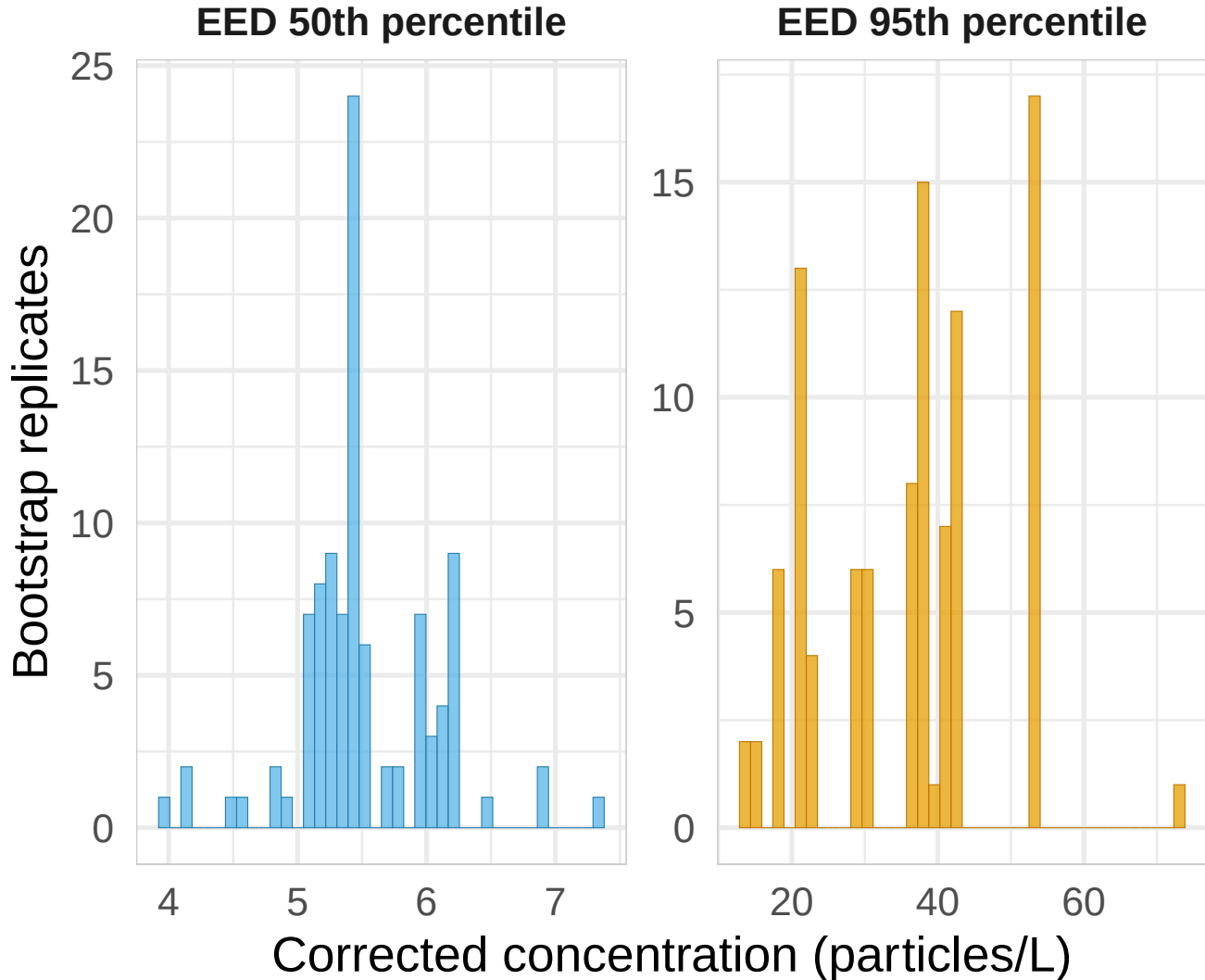
# Distribution of particle width-to-length ratio





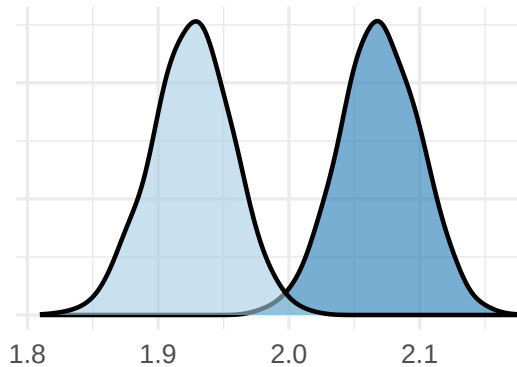
# Bootstrap uncertainty in Environmental Ex

River water — n = 100 bootstrap replicates of site-

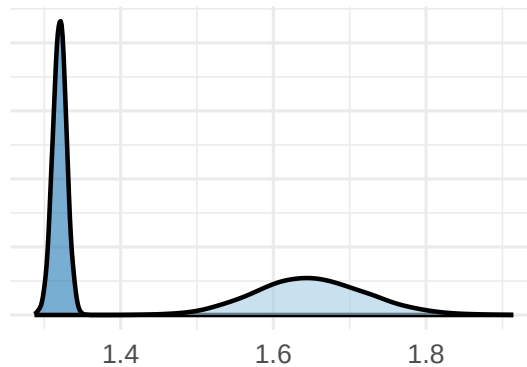


Correction factor (CF) propagated via Monte Carlo; each replicate resamples sites with replacement.

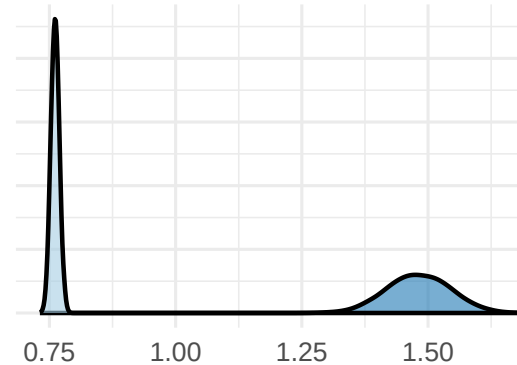
### Length Alpha



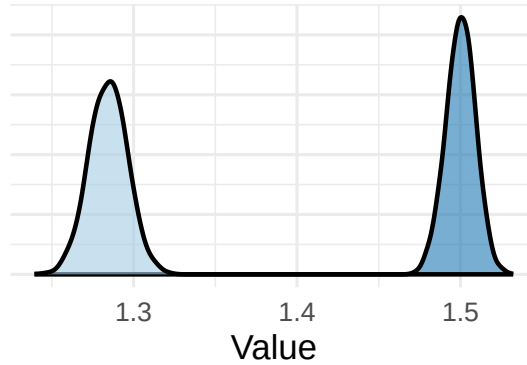
### Mass Alpha



### Volume Alpha



### Surface Area Alpha

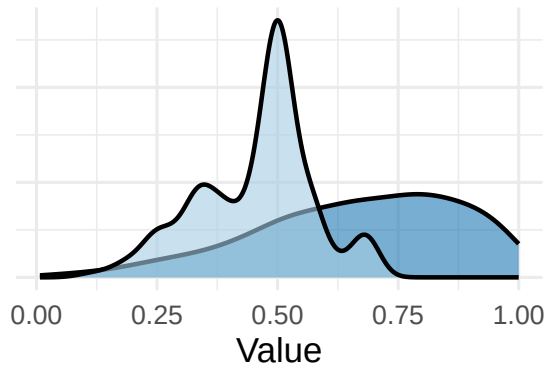


Group

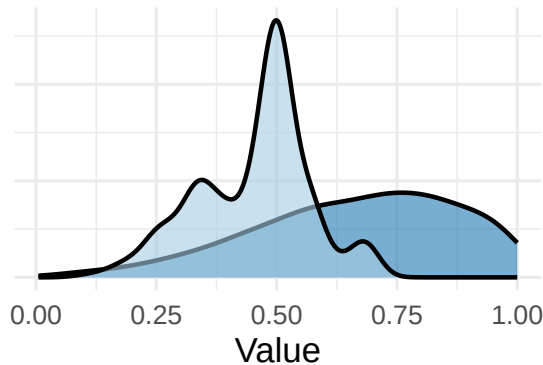
Marine

Freshwater

### Length to Width Ratio

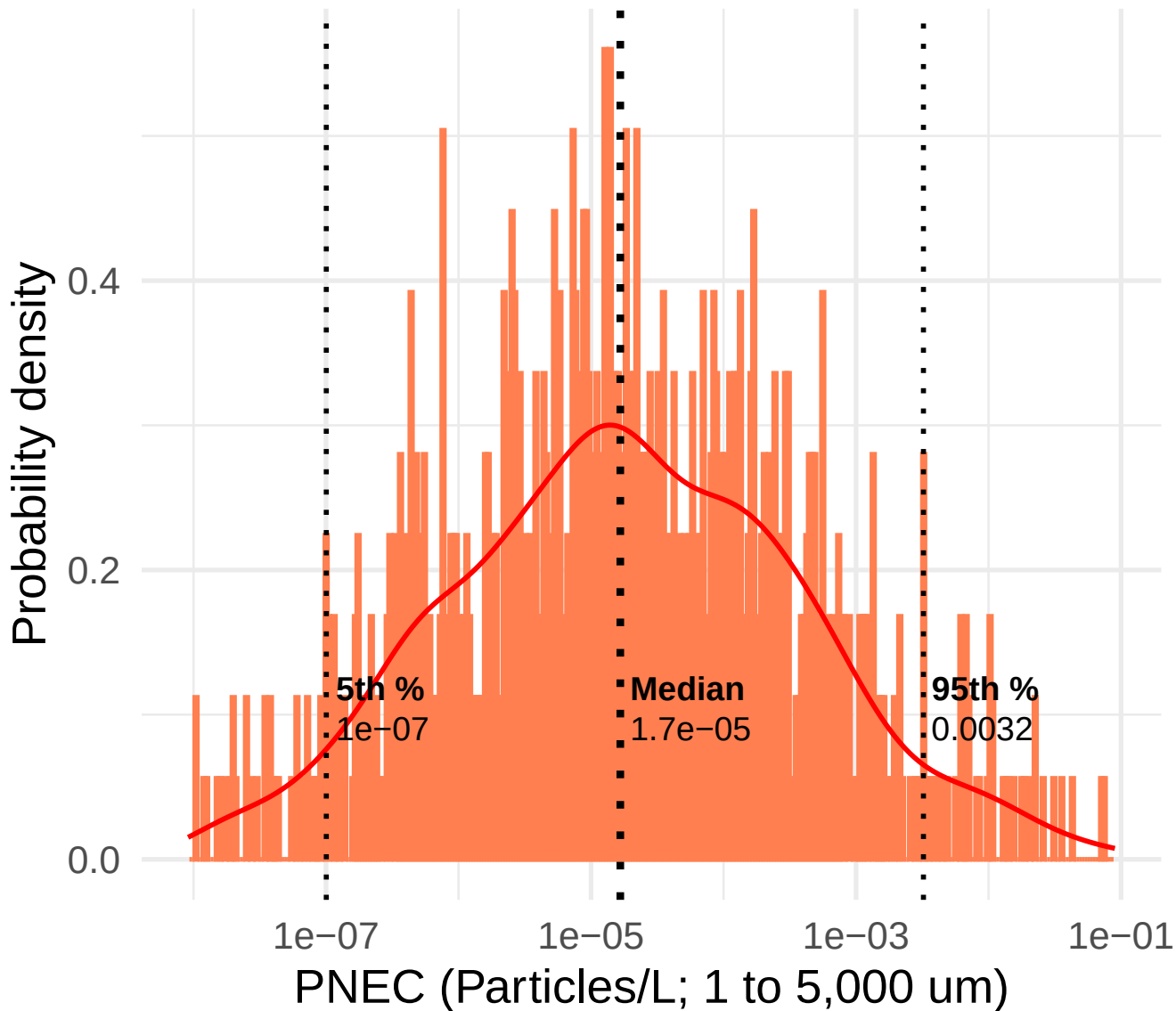


### Height to Width Ratio



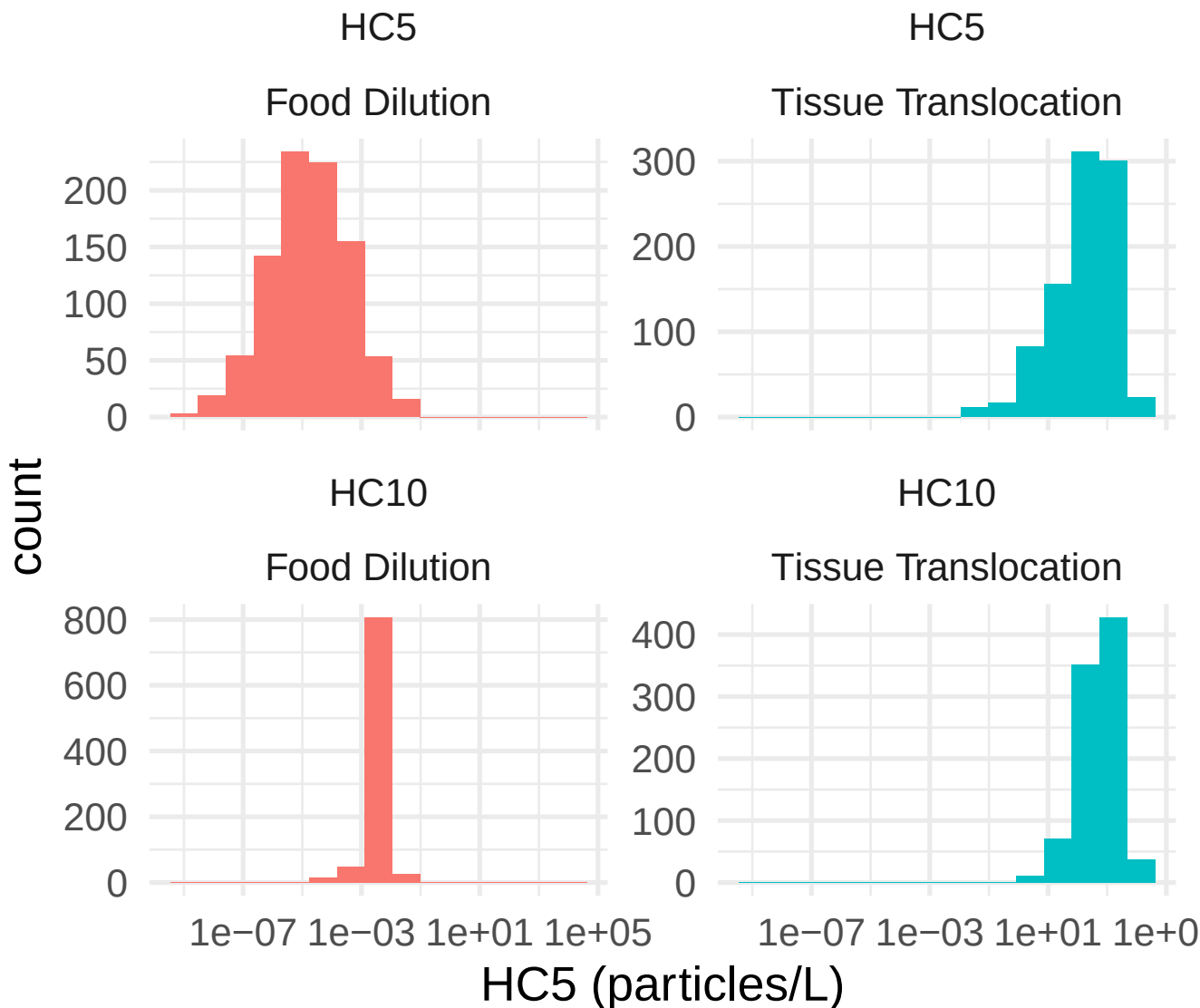
# Tier 3 (HC5)

River water

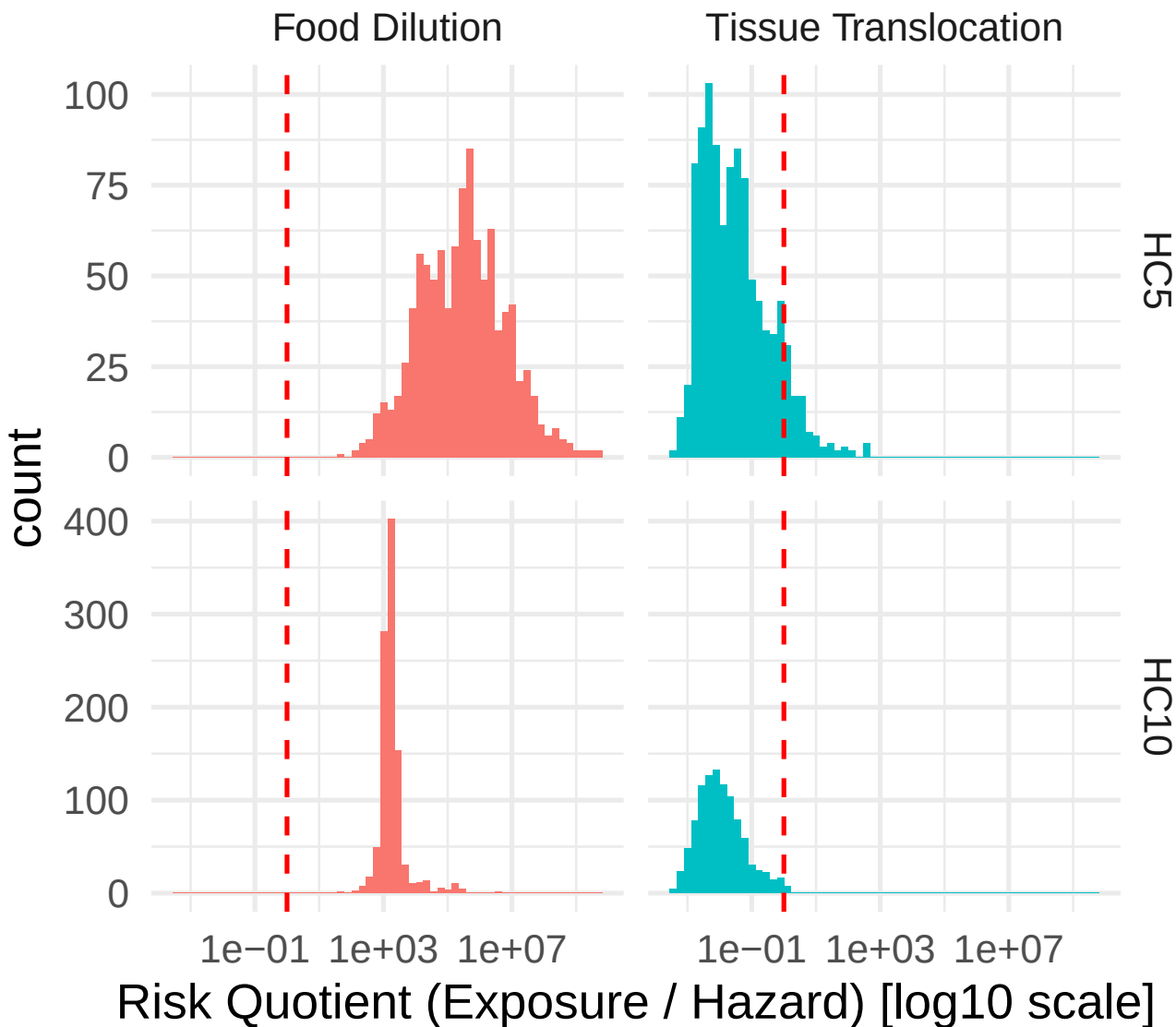


# Hazard threshold distribution

River water

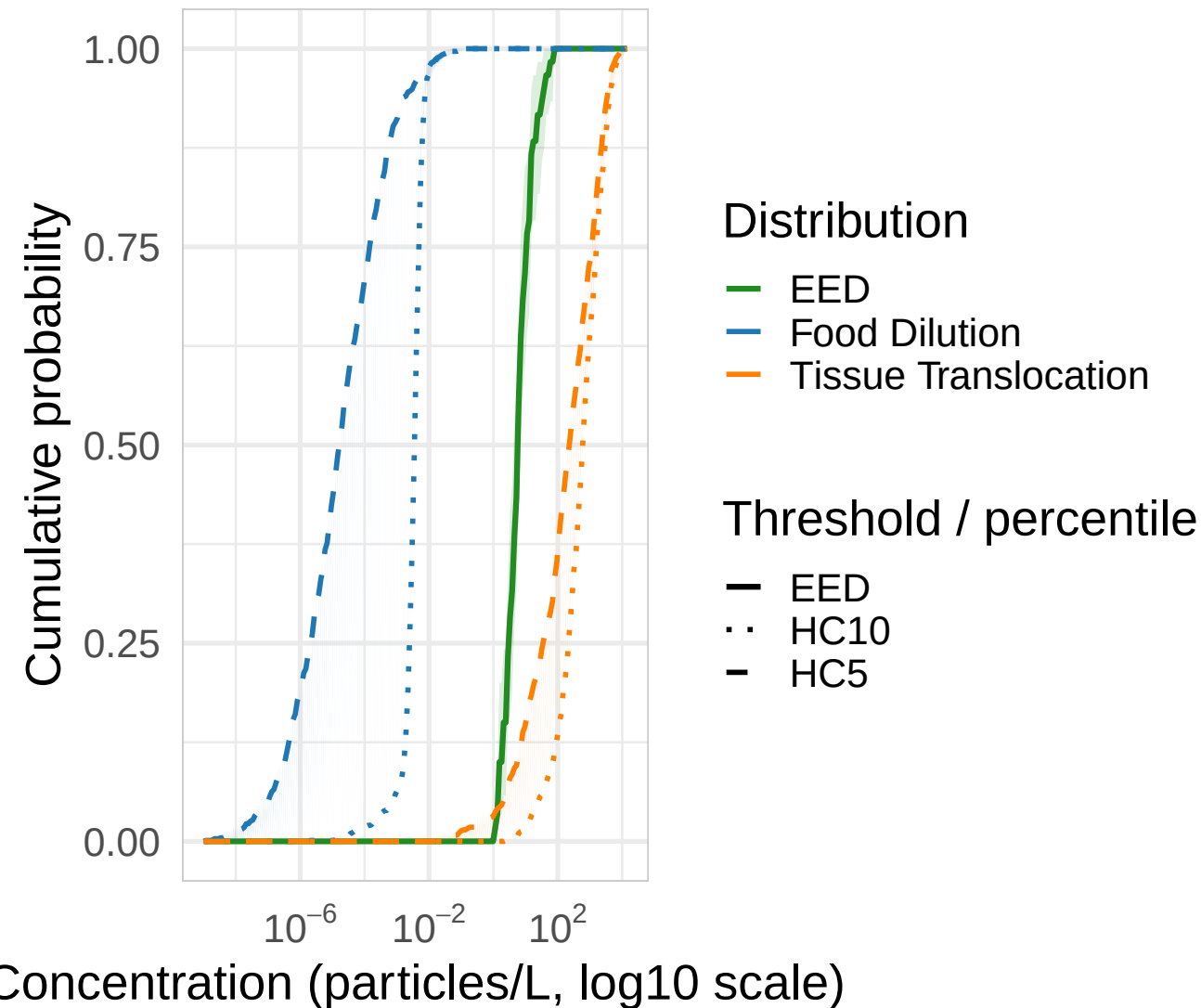


# Monte Carlo risk quotient distributions by River water



# Overlap of Exposure and Hazard distribution

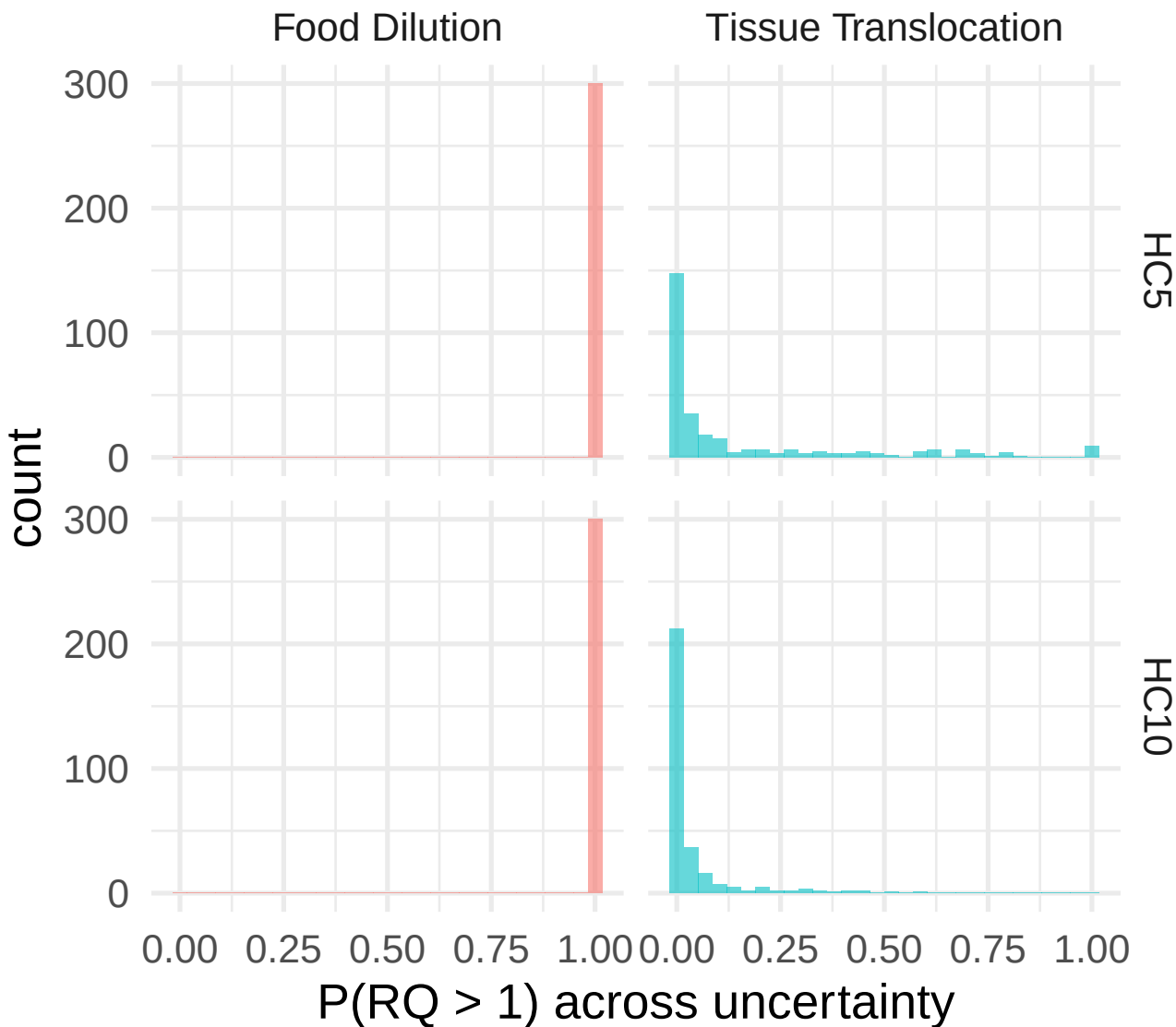
River water — EED = Environmental Exposure Distribution



Shaded bands = bootstrap 95% CI (n = 100 replicates). Log10 x-axis.

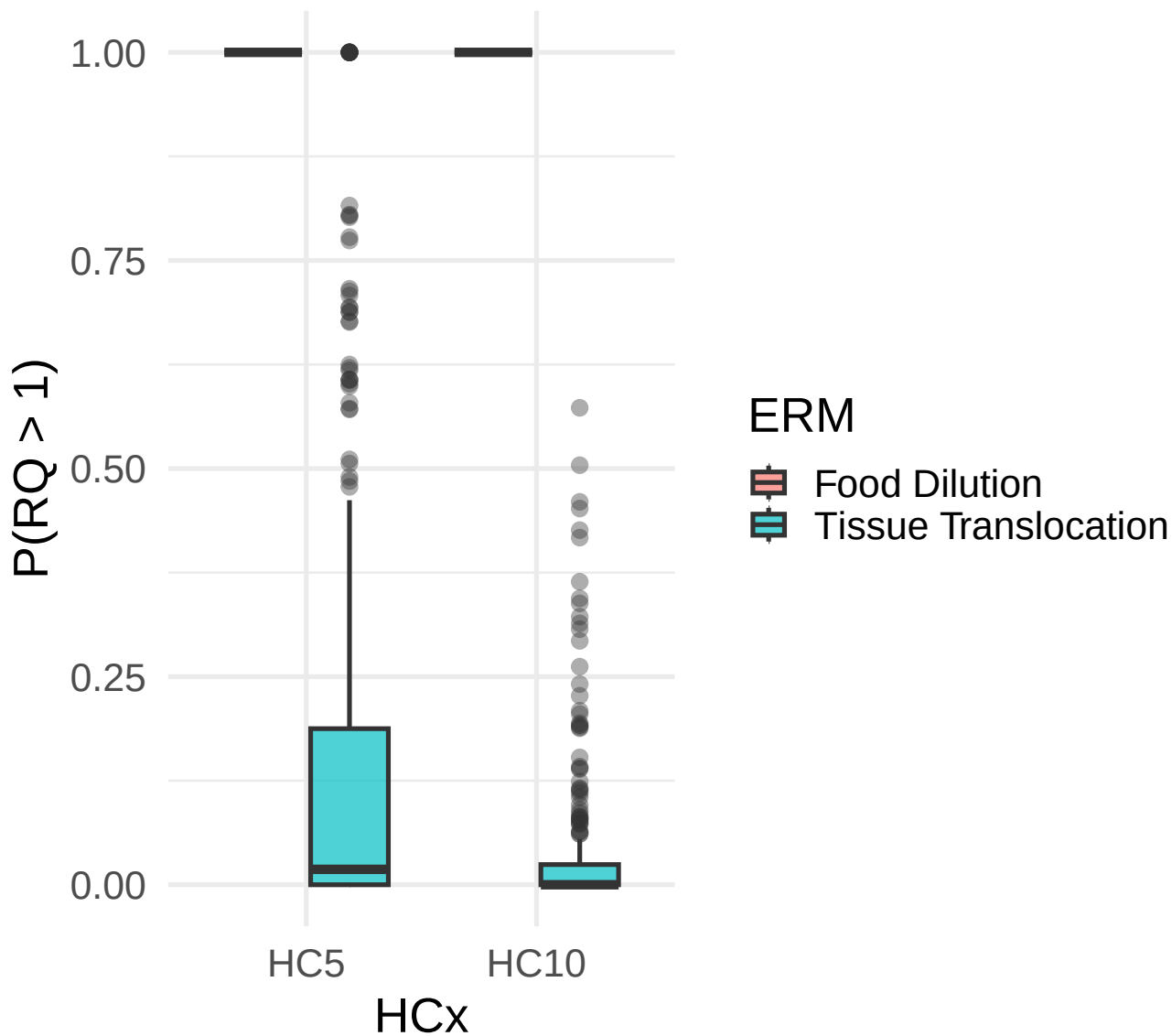
# MC2D uncertainty distribution for exceeded

## River water



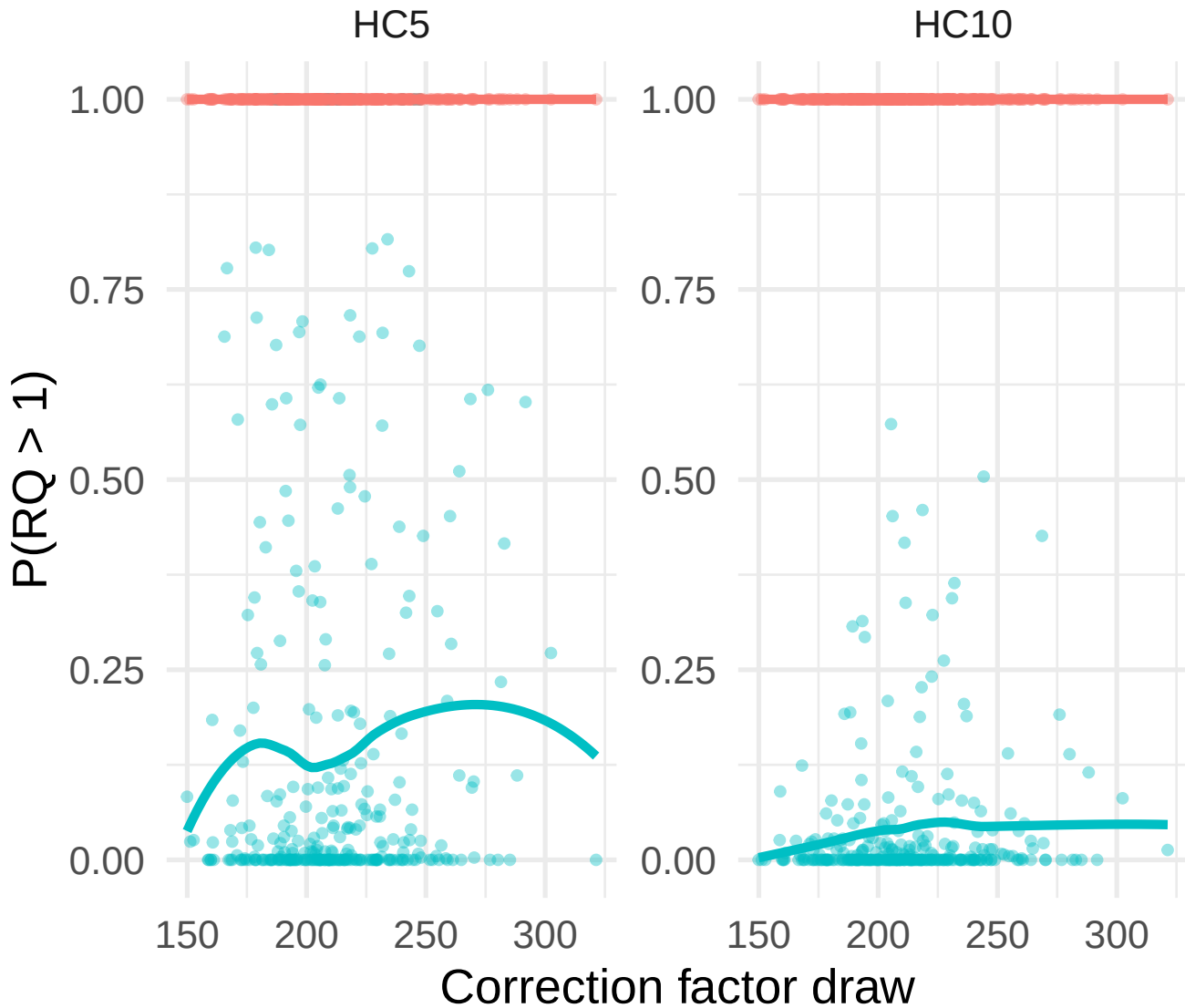
# MC2D uncertainty spread in exceedance

River water





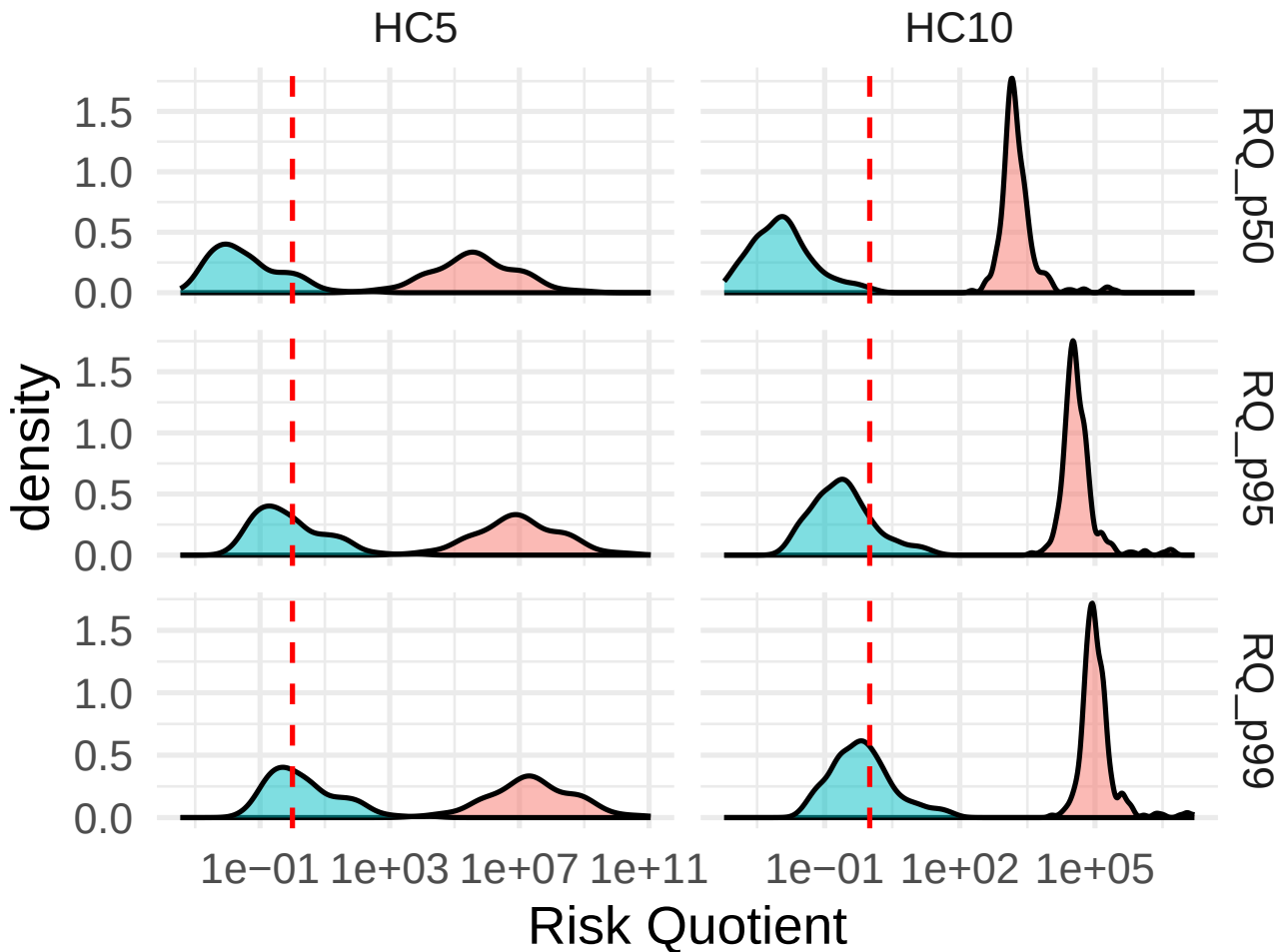
# Relationship between correction factor and River water



# MC2D uncertainty distribution for RQ sum

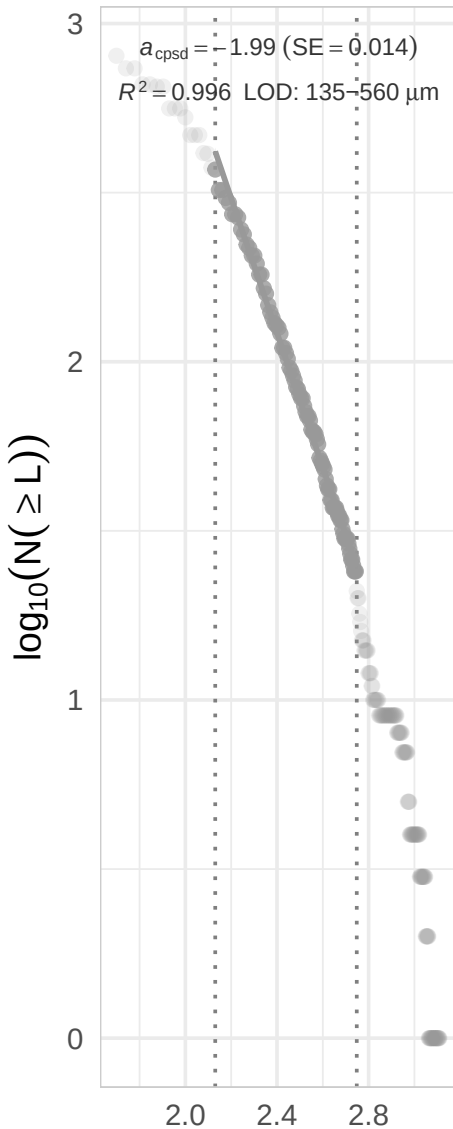
River water

ERM ■ Food Dilution ■ Tissue Translocation

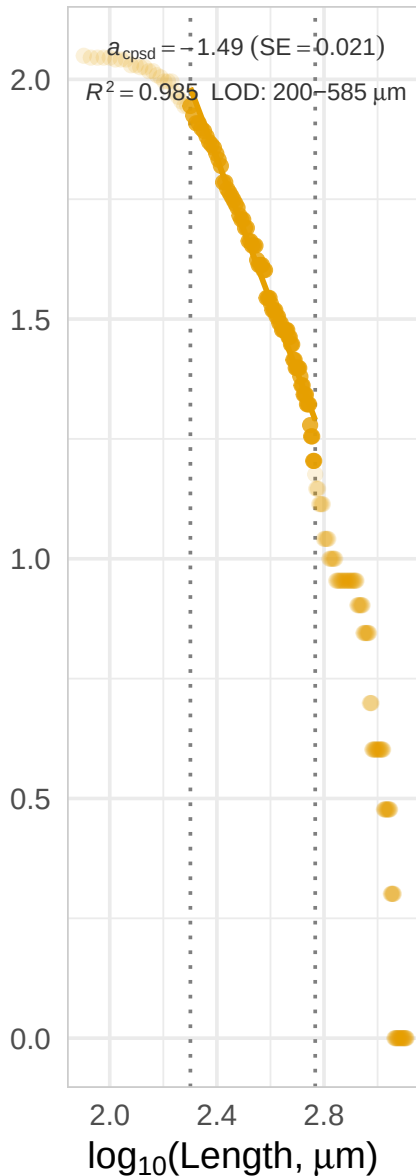


# Sediment C-PSD fits by shape

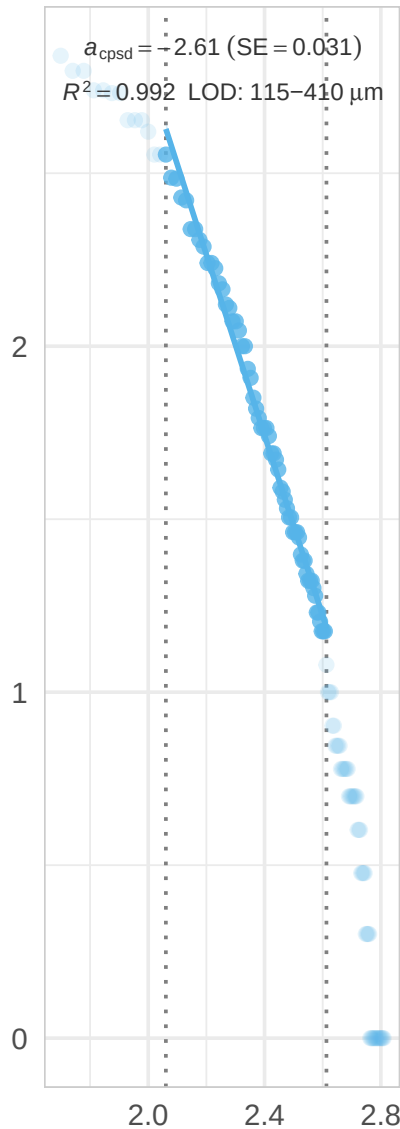
**all**



**fiber**

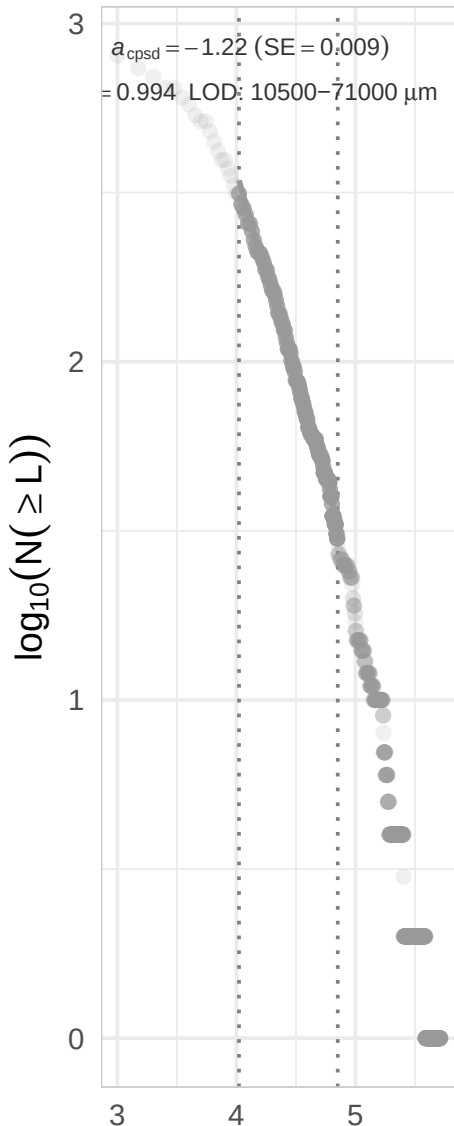


**fragment**

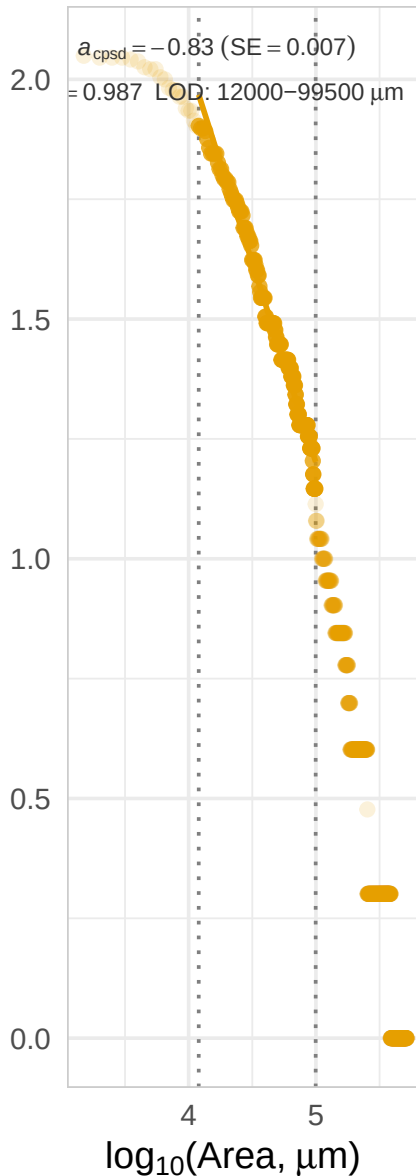


# Sediment C-PSD fits by shape

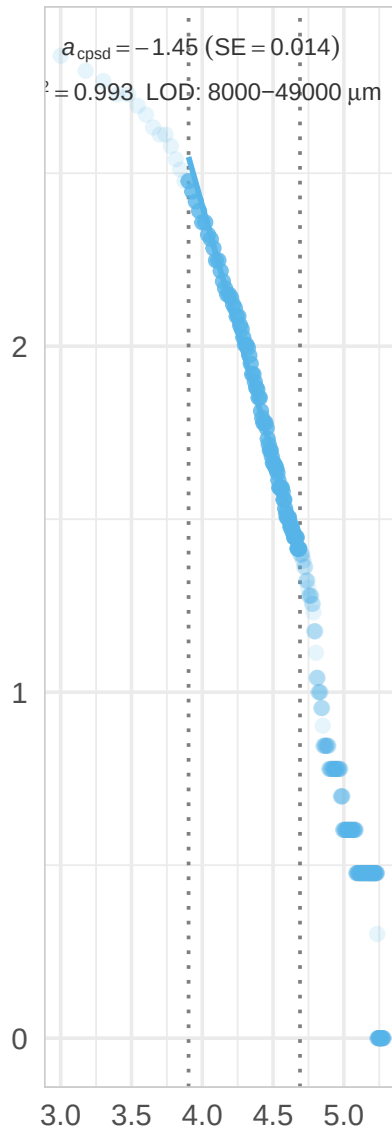
**all**



**fiber**

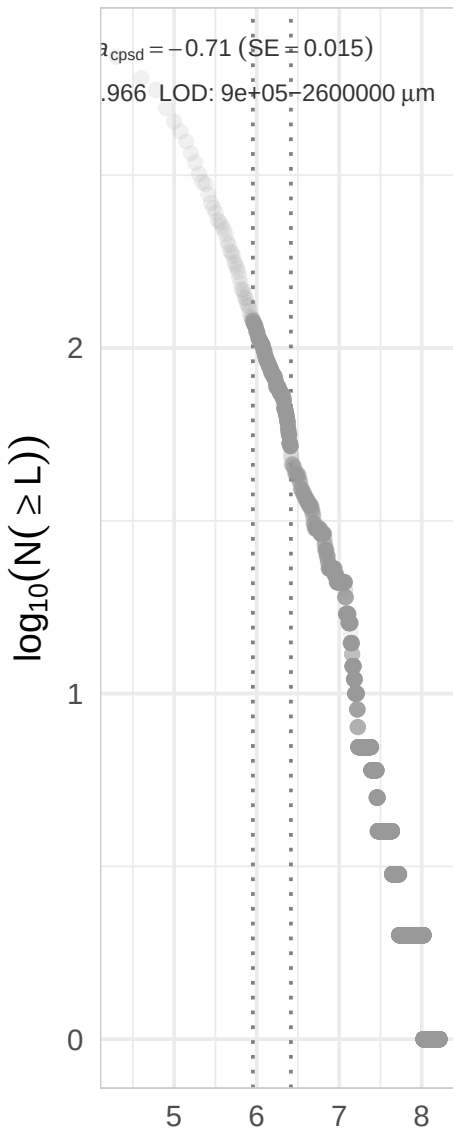


**fragment**

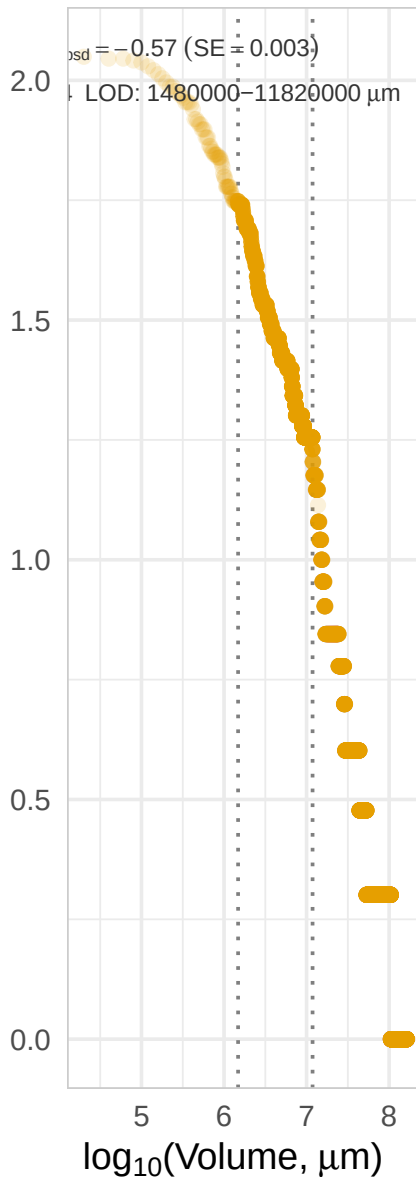


# Sediment C-PSD fits by shape

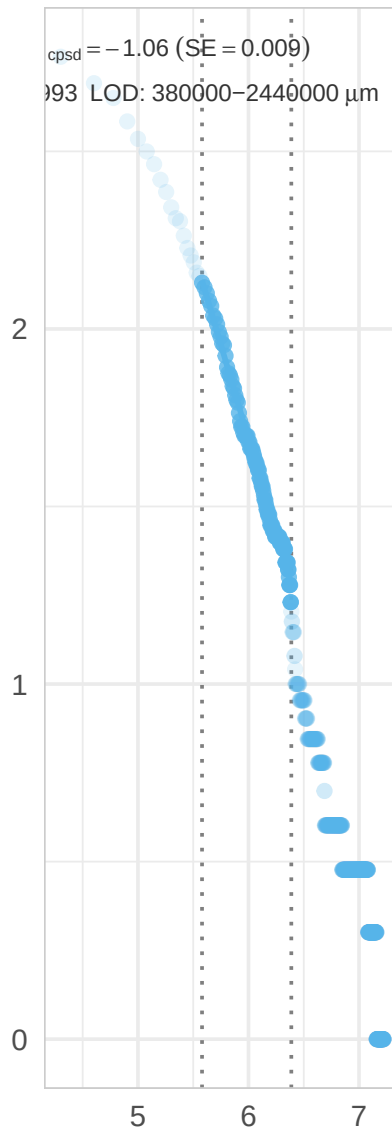
**all**



**fiber**



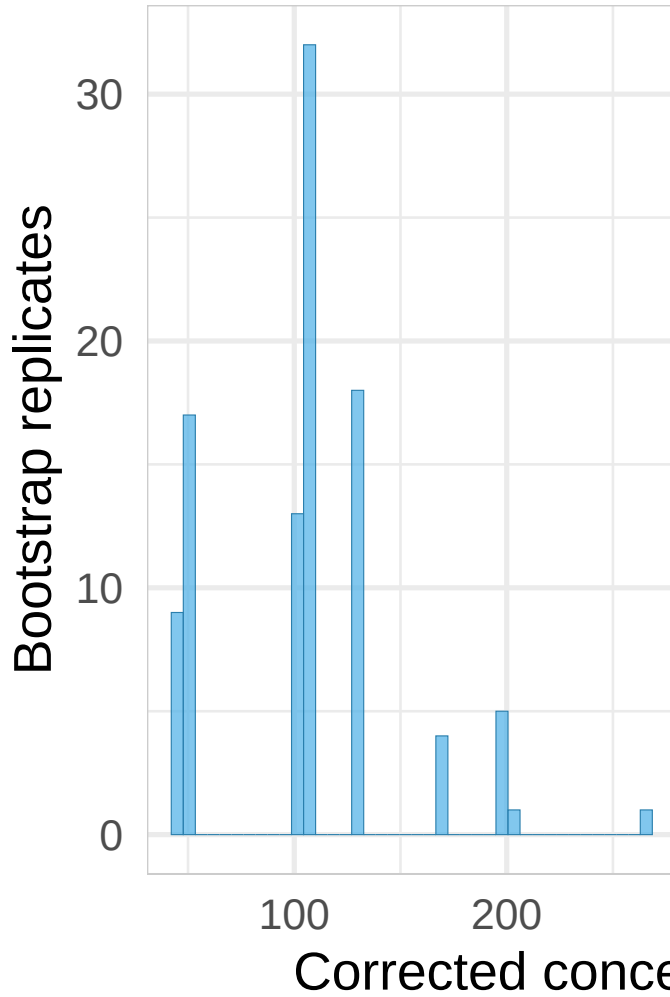
**fragment**



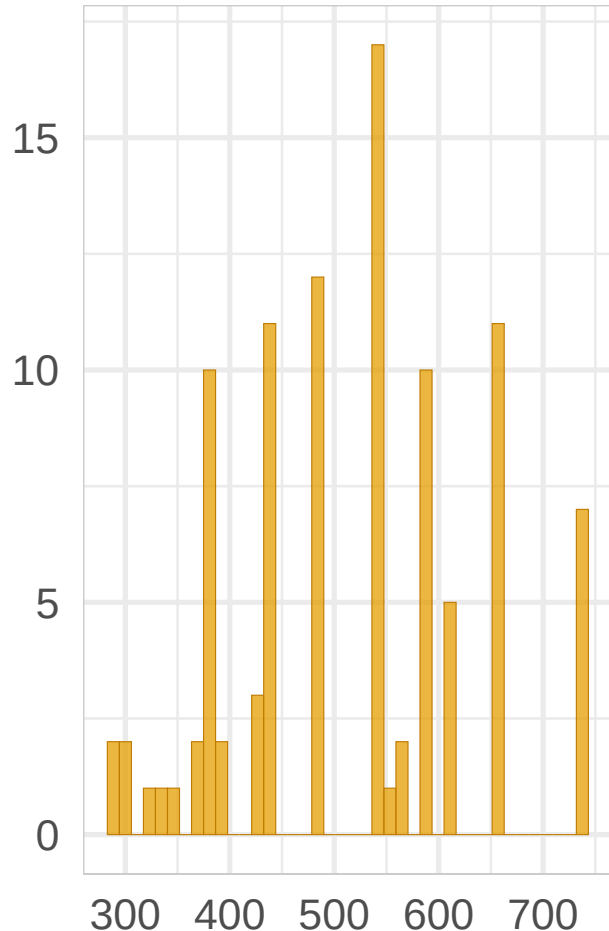
# Bootstrap uncertainty in Environmental Ex

Sediment —  $n = 100$  bootstrap replicates of site-le

**EED 50th percentile**

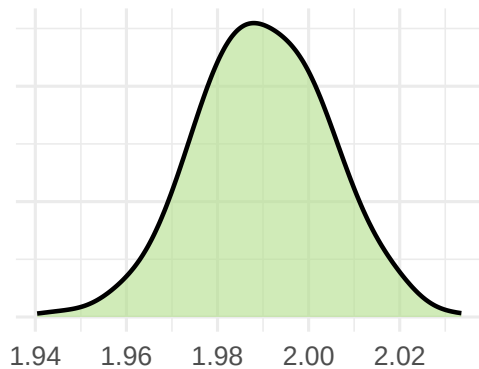


**EED 95th percentile**

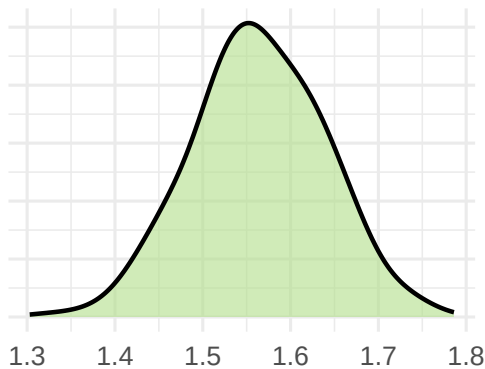


Correction factor (CF) propagated via Monte Carlo; each replicate resamples sites with replacement.

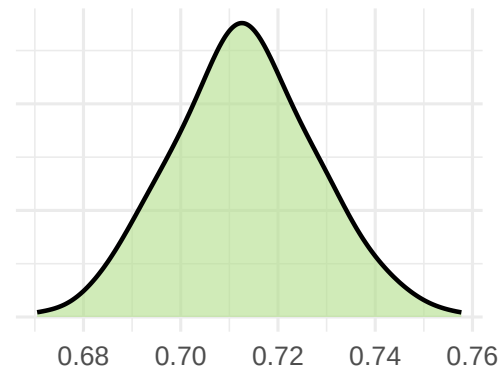
**Length Alpha**



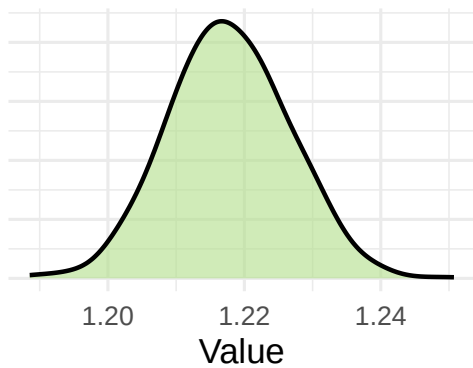
**Mass Alpha**



**Volume Alpha**



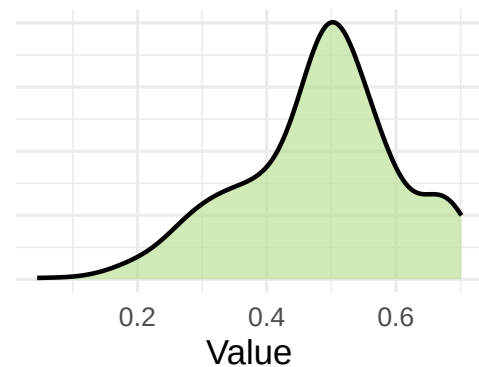
**Surface Area Alpha**



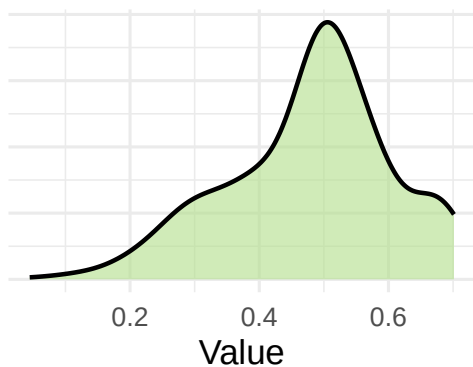
Group

 Freshwater Sediment

**Length to Width Ratio**

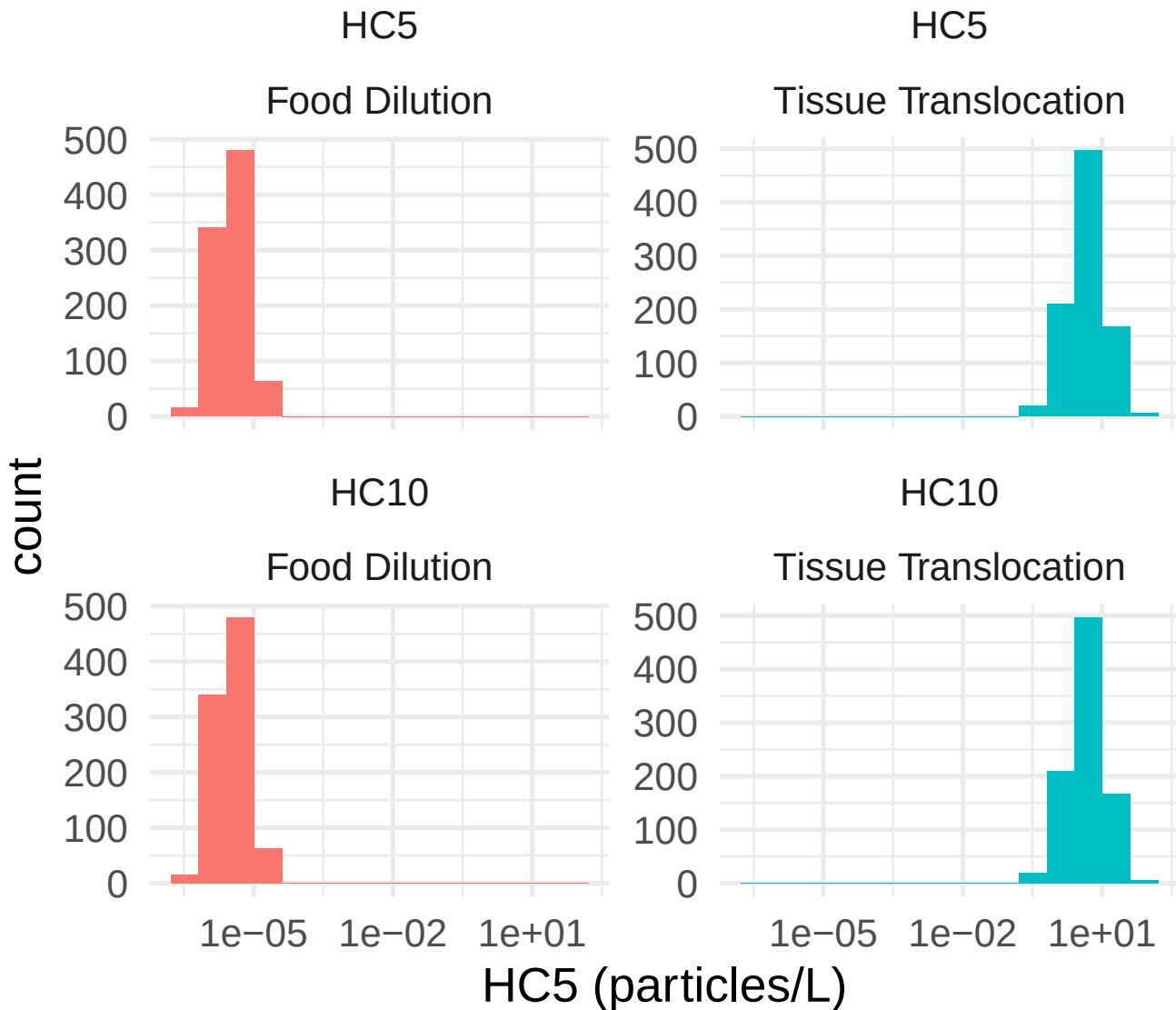


**Height to Width Ratio**



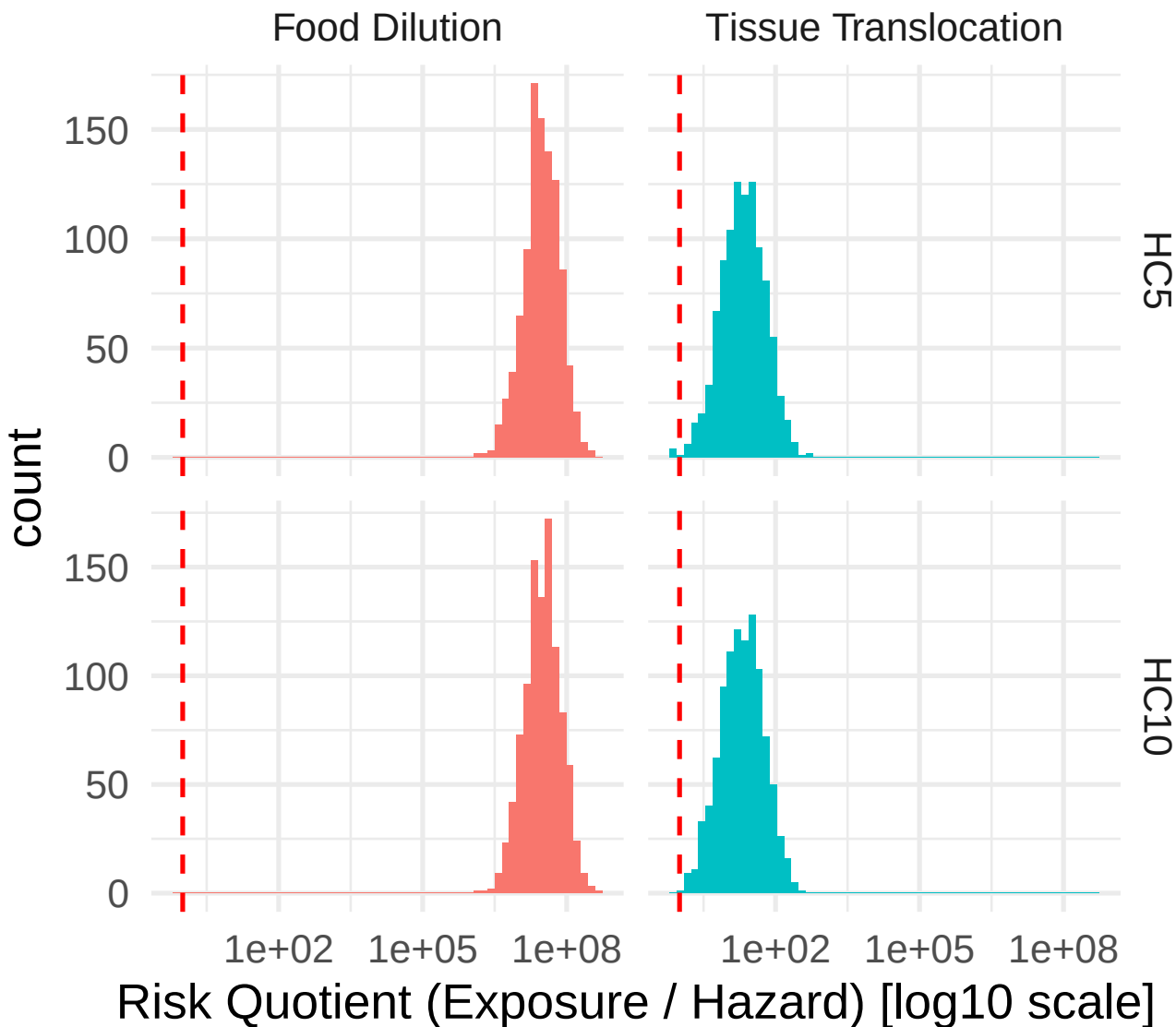
# Hazard threshold distribution

Sediment



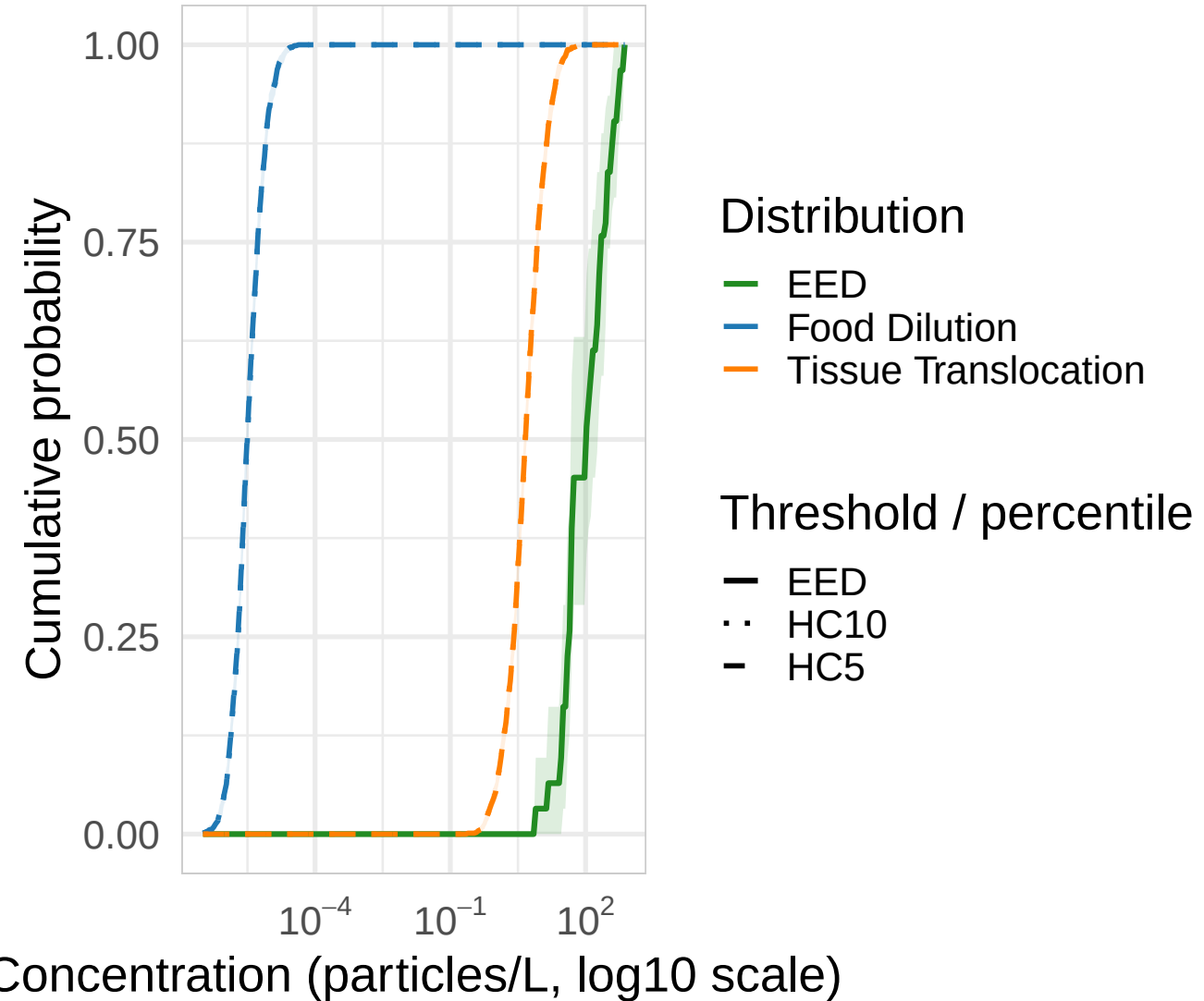


# Monte Carlo risk quotient distributions by Sediment



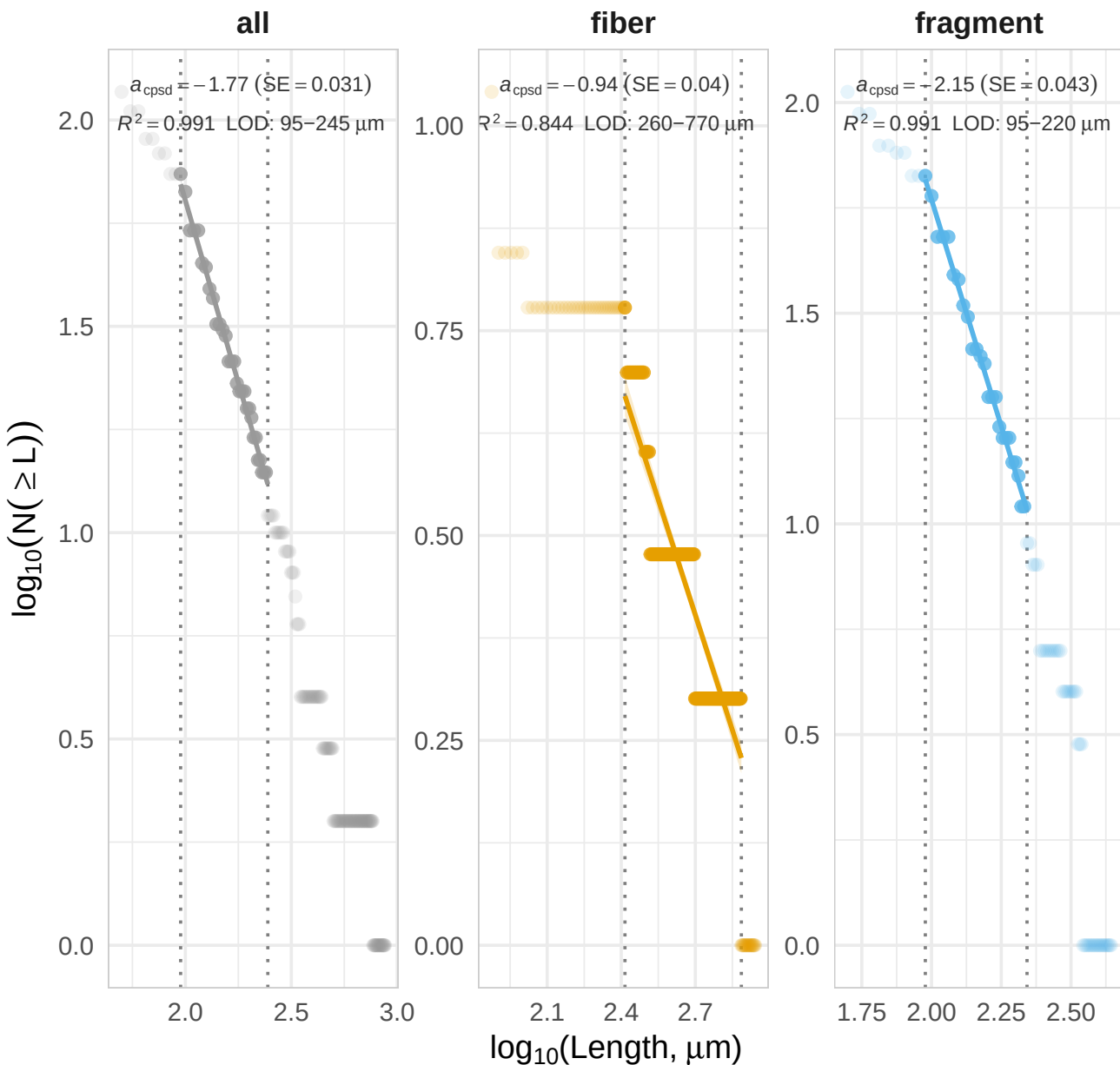
# Overlap of Exposure and Hazard distributions

Sediment — EED = Environmental Exposure Distribution

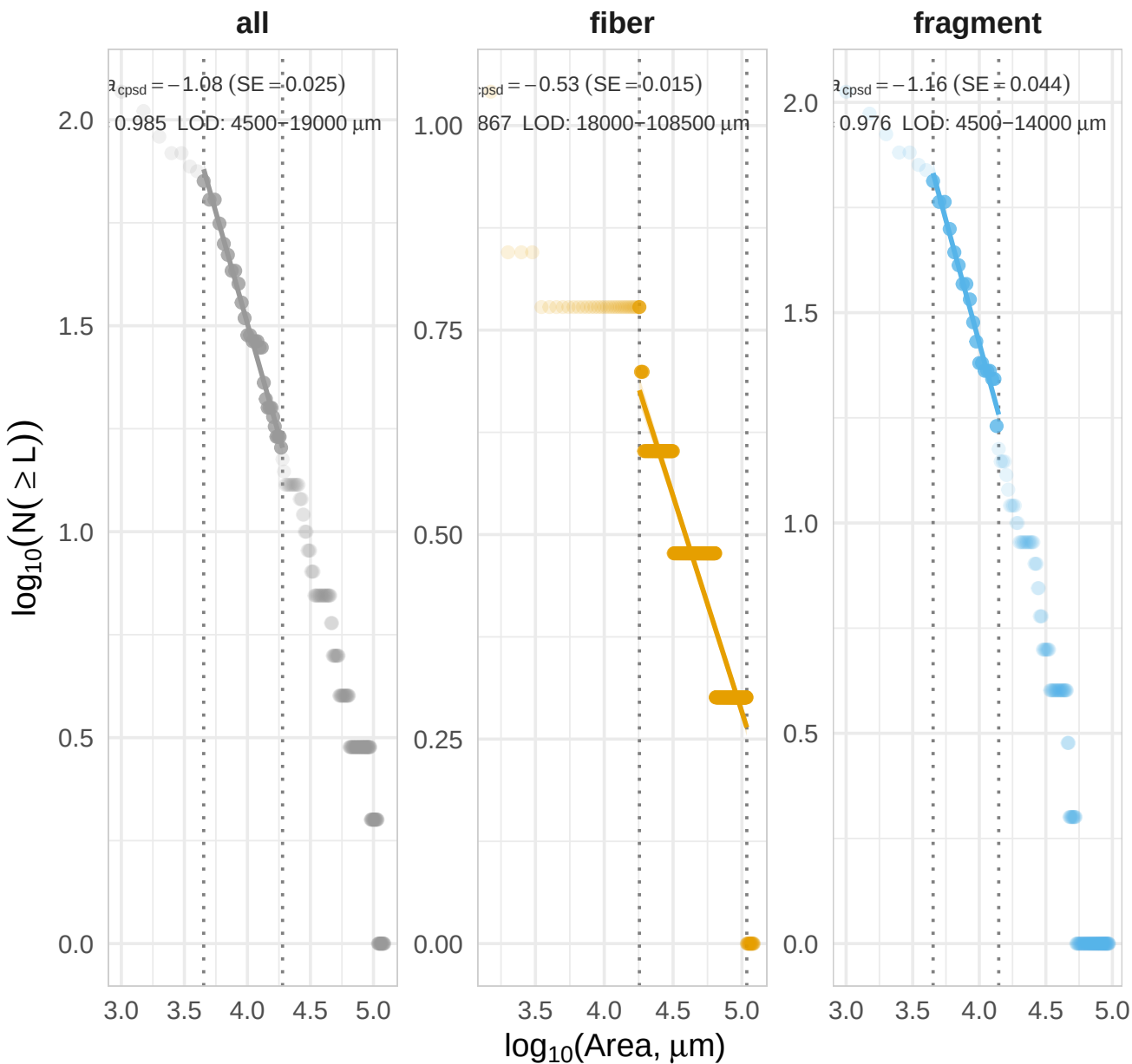


Shaded bands = bootstrap 95% CI (n = 100 replicates). Log10 x-axis.

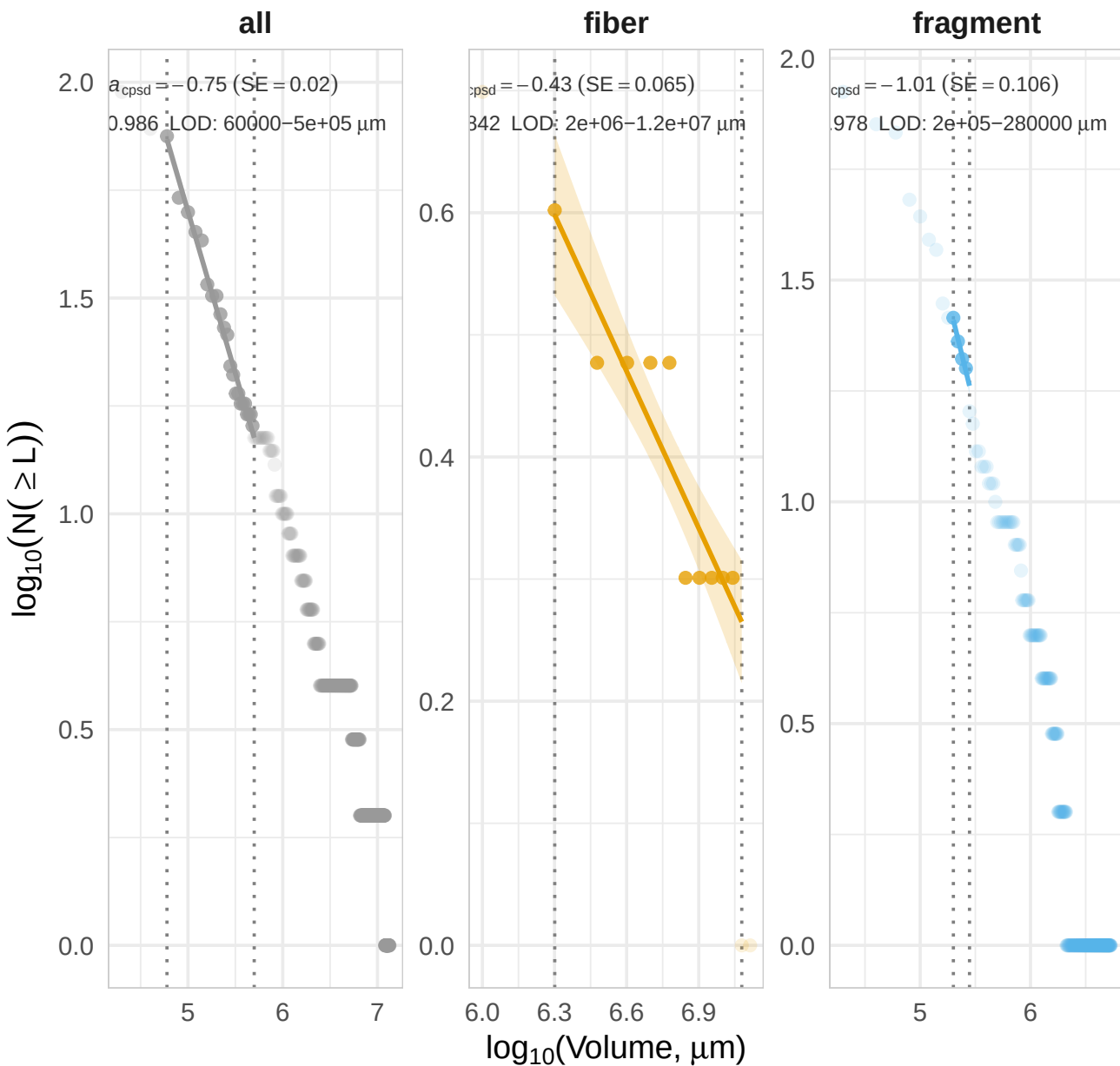
# Ocean water C-PSD fits by shape



# Ocean water C-PSD fits by shape



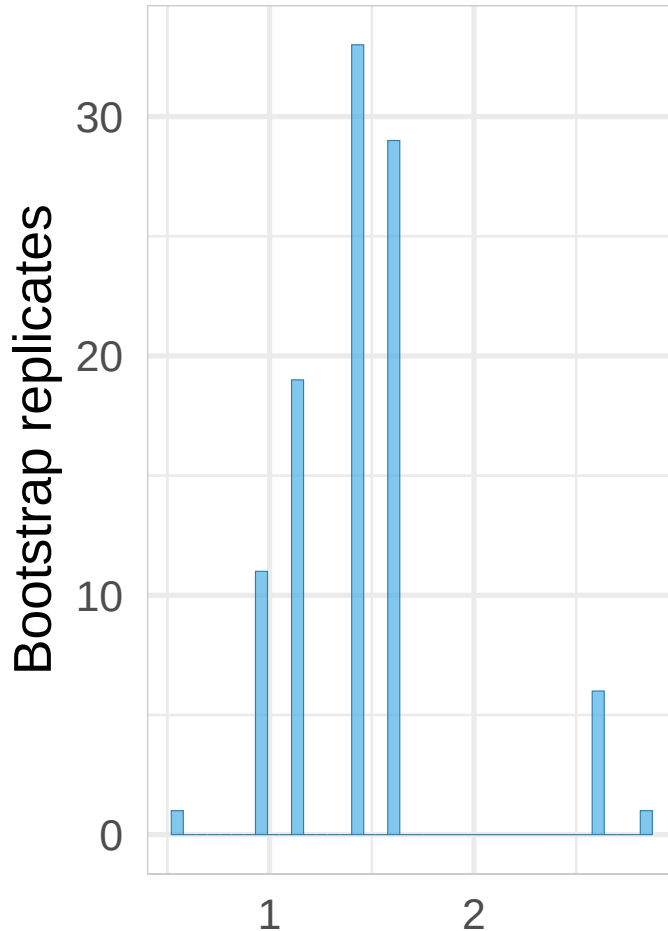
# Ocean water C-PSD fits by shape



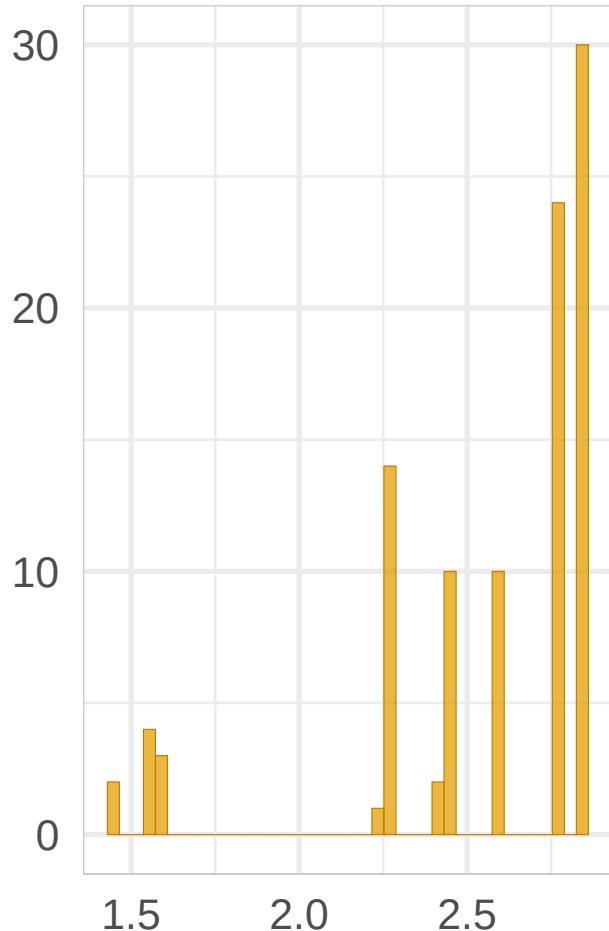
# Bootstrap uncertainty in Environmental Ex

Ocean water — n = 100 bootstrap replicates of site

**EED 50th percentile**



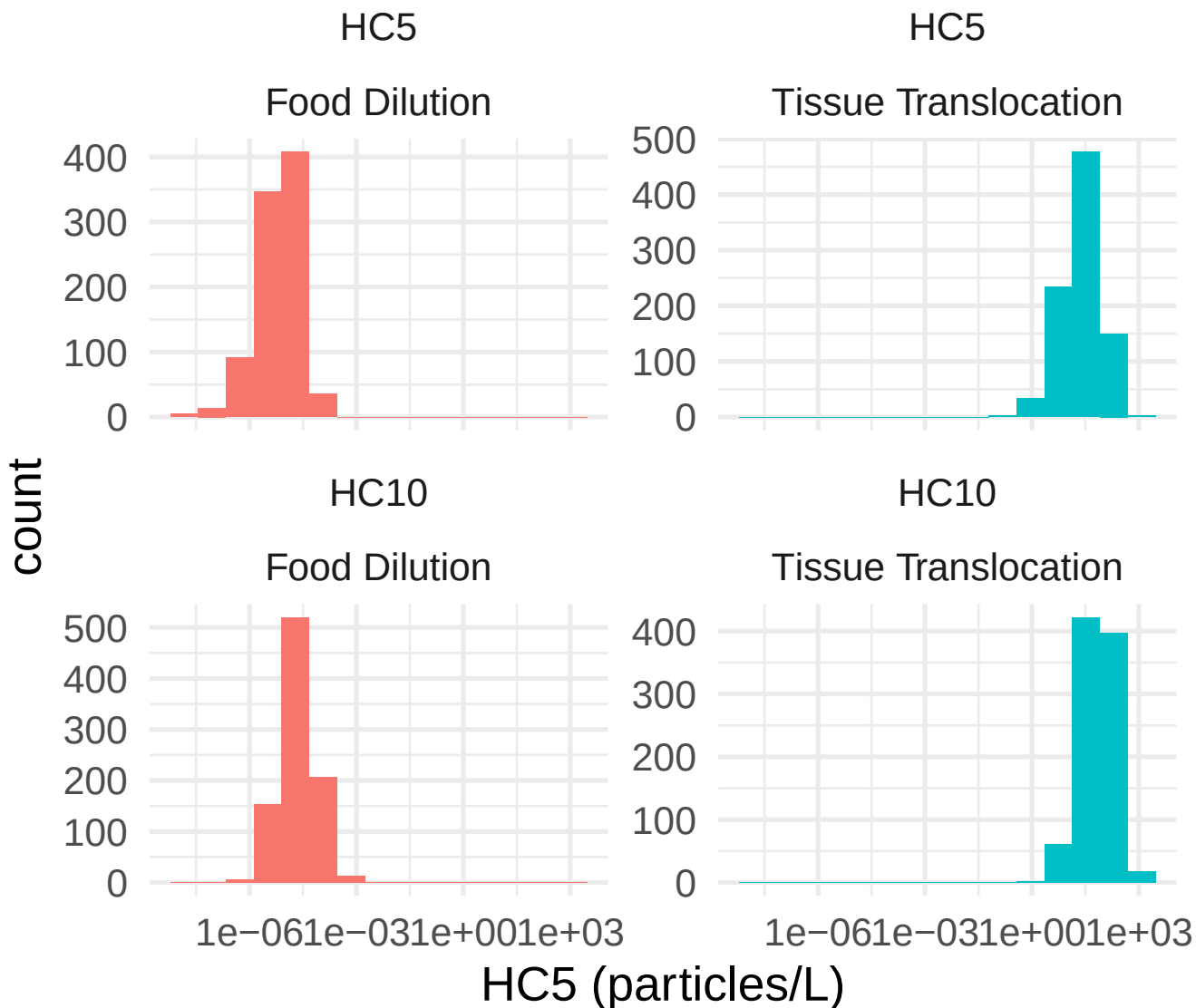
**EED 95th percentile**



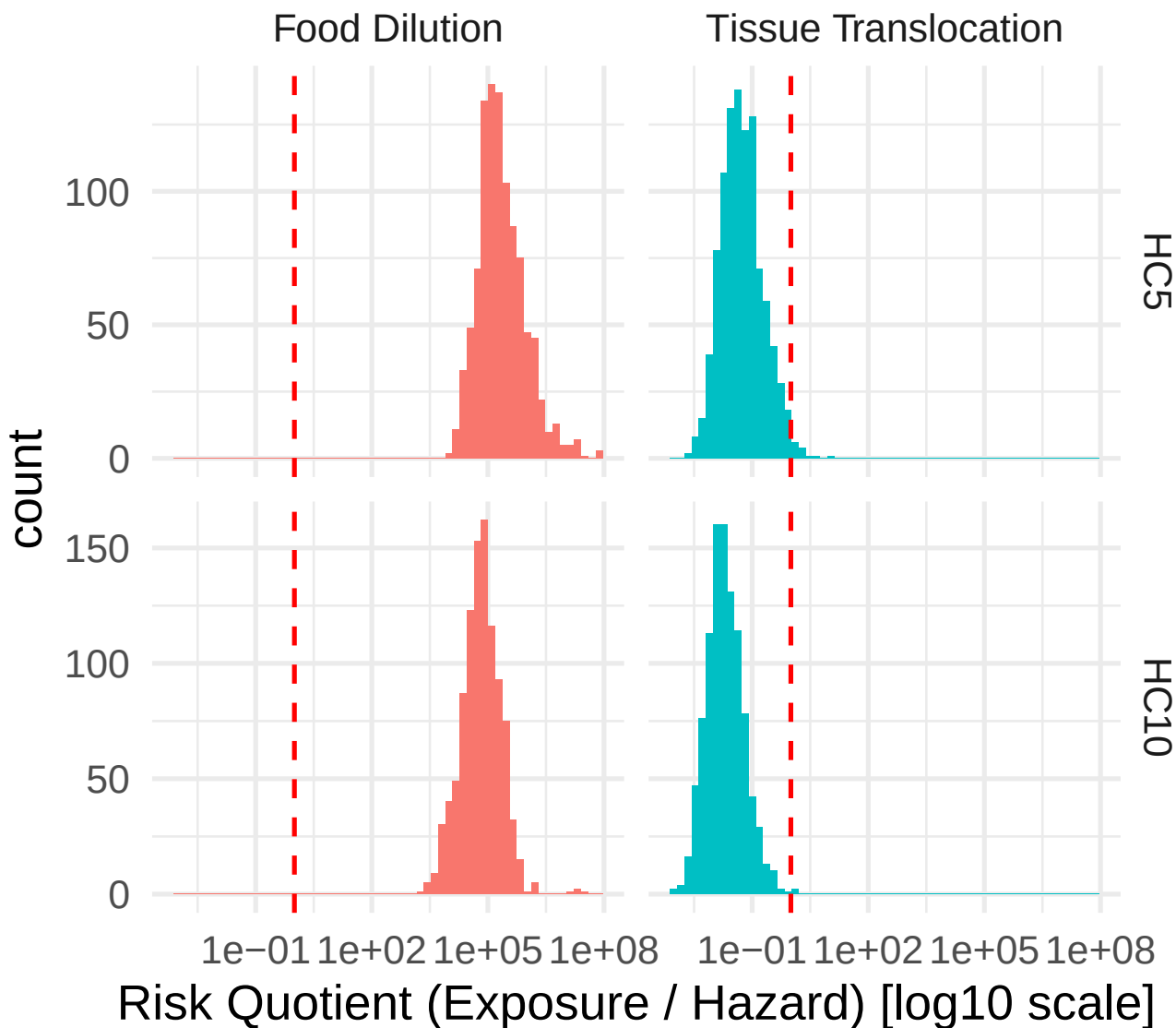
Correction factor (CF) propagated via Monte Carlo; each replicate resamples sites with replacement.

# Hazard threshold distribution

Ocean water



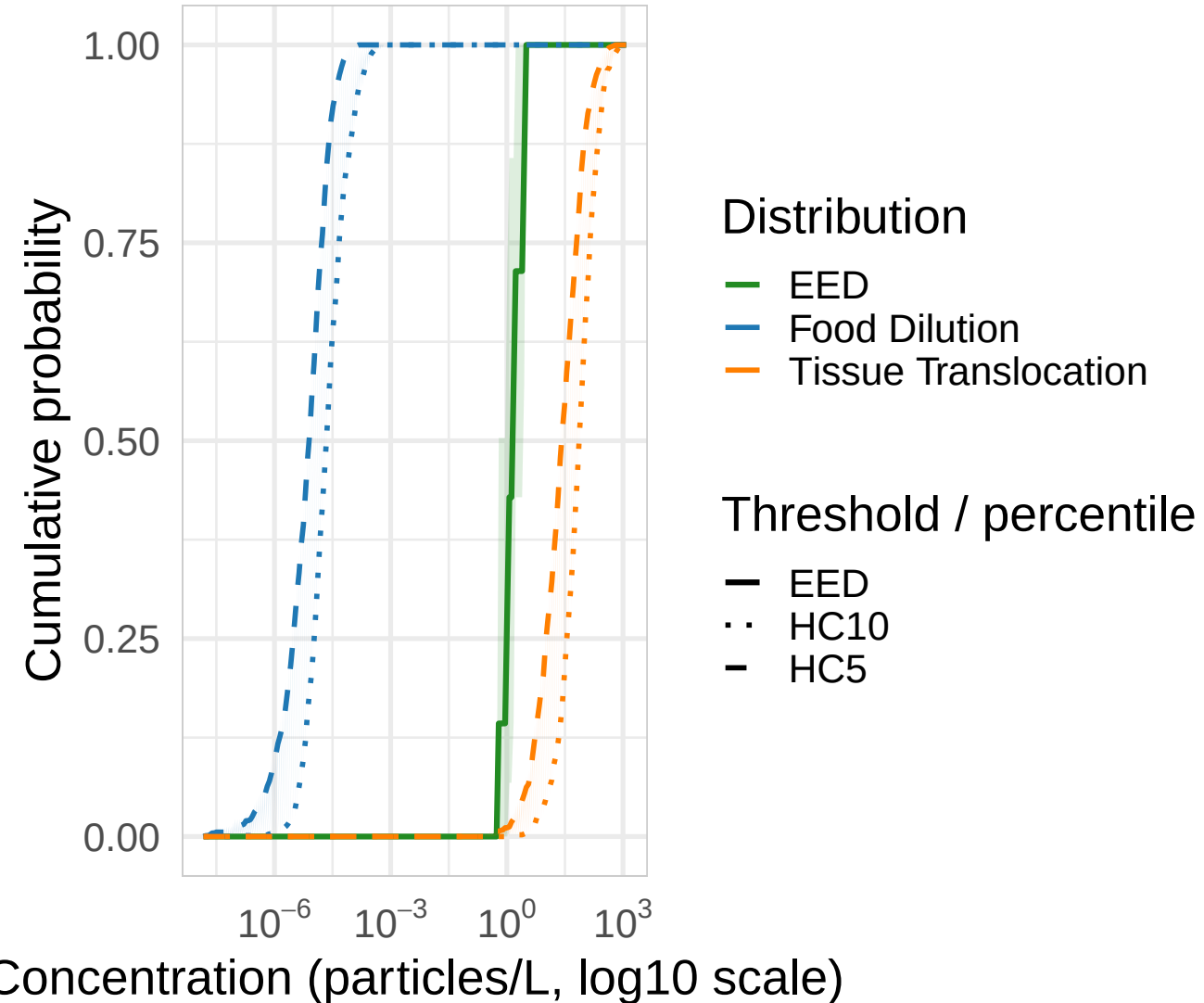
# Monte Carlo risk quotient distributions by Ocean water





# Overlap of Exposure and Hazard distribution

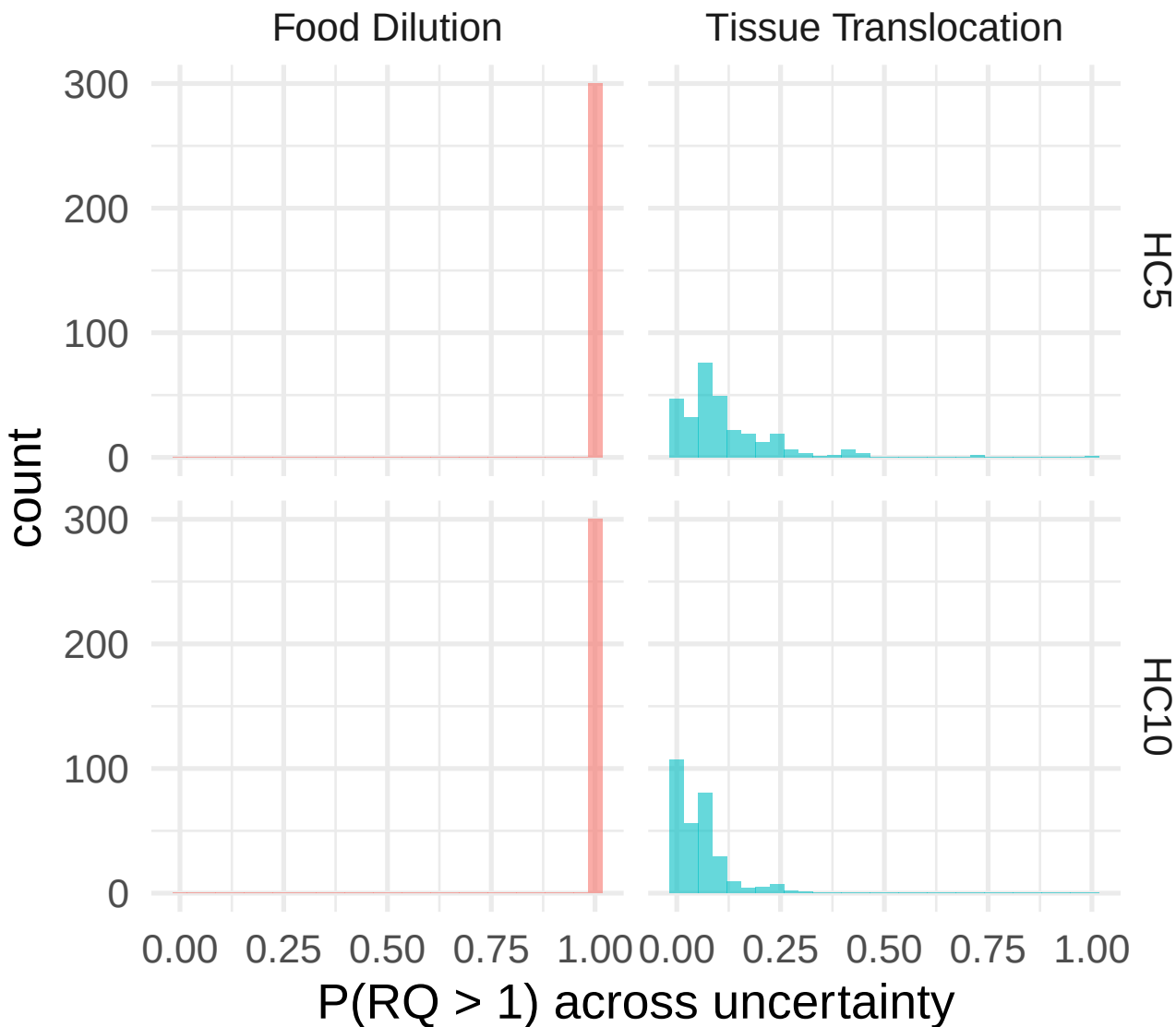
Ocean water — EED = Environmental Exposure



Shaded bands = bootstrap 95% CI (n = 100 replicates). Log10 x-axis.

# MC2D uncertainty distribution for exceeded

## Ocean water



# 1D Monte Carlo risk quotient comparison across matrix

Rows = matrix, columns = exposure route model (ERM). Dashed red line indicates a threshold value.

